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Department of Industrial Engineering

IE 3511
Systems Simulation

Group 3

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1. Executive Summary

This study developed a discrete-event simulation model for the departure process of Regional Airport (SRA) using Arena. The main purpose of the project was to evaluate the current passenger flow, identify the main causes of missed flights, and compare possible improvement scenarios under a fixed investment budget of \$1 million. The model covers the passenger journey from airport arrival to the assigned gate. It includes transportation mode, check-in, baggage drop, security screening, passenger-tray synchronization, terminal walking, gate routing, and missed flight recording. In the model, passengers are considered late if they cannot reach the required point before the gate closing time, which is assumed to be 10 minutes before flight departure.

The input modeling stage was completed by analyzing the provided data sets in Arena Input Analyzer. Before selecting distributions, the IID properties of the data were checked using visual and statistical methods such as scatter plots, autocorrelation plots, and subgroup comparisons. Some processes, especially security-related activities, were represented with theoretical distributions such as triangular and Weibull distributions. However, larger and more irregular data sets, such as check-in and bag drop processes, were modeled using empirical distributions to preserve the original behavior of the data more realistically. This approach helped improve the credibility of the simulation model.

The base model was verified and validated before testing improvement scenarios. Verification was performed by checking Arena logic, passenger routing, queues, counters, and animation behavior. Validation was mainly based on comparing the base model output with the expected missed flight behavior in the project description. The base scenario produced a missed flight rate of 10.76%, which was accepted as a reasonable representation of the current system. This value was then used as the reference point for evaluating all improvement alternatives.

Six improvement scenarios were tested. Scenario 1, which added self-service kiosks and budget check-in counters, reduced the missed flight rate to 5.54%. This was the strongest individual improvement and also the most cost-effective option. Scenario 2, which expanded standard security capacity, did not improve the system and resulted in a missed flight rate of 10.99%. Scenario 3, which increased bag drop capacity, reduced the missed flight rate to 9.27%. Scenario 4, which added moving sidewalks, produced the same missed flight rate as the base case, 10.76%. Scenario 5, which introduced an advanced security lane, reduced the missed flight rate to 9.11%, but its improvement was limited compared with its high cost.

The best overall performance was obtained in Scenario 6, the integrated final strategy. This scenario combined several improvements, including additional self-service kiosks, improved bag drop capacity, advanced security capacity, and extra divesting and revesting capacity. It reduced the missed flight rate from 10.76% to 4.29%, which corresponds to a 60.1% reduction compared with the base case. This result shows that the missed flight problem is not caused by only one station. Instead, delays accumulate across several connected stages of the passenger journey.

Based on the simulation results, Scenario 1 is recommended as the most practical short-term solution because it provides a strong improvement with a relatively low investment cost. Scenario 6 is recommended as the strongest long-term strategy if airport management wants to achieve the maximum reduction in missed flights. Overall, the study shows that SRA should focus on reducing early-stage delays and improving flow continuity across the system instead of improving only one isolated process. The Arena simulation model provides a useful decision-support tool for comparing operational alternatives before making real investment decisions.

2. Introduction & Objective

In recent years, Regional Airport (SRA) has experienced a steady increase in passenger demand due to economic growth in the surrounding region. While this growth is positive, it has also created operational challenges for the airport. The existing terminal layout and service capacity are no longer sufficient to handle the increased passenger volume efficiently, leading to congestion and longer waiting times, especially during peak periods.

To address these issues, SRA has been allocated a fixed budget of \$1 million. The main objective of this study is to determine how this budget can be used in the most effective way to improve overall system performance. This includes improving key processes as well as evaluating different investment alternatives. The study focuses on reducing waiting times, minimizing queue lengths at critical service points, and lowering the number of passengers who miss their flights.

The analysis focuses on the departure process, starting from passenger arrival at the airport and ending when passengers either reach their assigned departure gates or leave the system. This includes check-in, baggage drop, and security screening. The security process is modeled as a multi-step system including ID check, divesting, X-ray screening, and possible additional procedures such as body search and manual bag inspection, as well as passenger movement within the terminal. While boarding itself is not modeled as a separate process, the model evaluates whether passengers are able to complete all required steps and reach their gates on time. Passengers who fail to do so are considered to have missed their flights and exit the system.

Since passenger arrivals and service times vary over time, a discrete-event simulation approach was used. The model captures how passengers move through the system under these varying conditions. It is based on the data provided in the project, including flight schedules, arrival patterns, and processing times, and it also considers differences in passenger characteristics such as baggage usage, check-in preferences, and routing decisions within the terminal.

The model is used to evaluate different improvement scenarios related to check-in, baggage handling, security operations, and passenger movement. Each scenario is assessed using performance measures such as waiting times, system utilization, and the number of missed

flights. These results are used to compare alternatives and support decision-making in selecting the most effective improvements within the given budget.

3. Group Organization and Responsibilities

Task	Responsible Group Member
Arena Input Analyzing, IID Verification, Scenario Development on Arena, Report (Executive Summary, Event Flow Charts, Input Modeling, Arena Model Design and Logic, Verification & Validation)	Rüstem Avcı
Report (Introduction & Objective, System Description, Assumptions & Simplifications, Conceptual Model, Experimentation & Scenarios, Results, Analysis & Discussion, Conclusion & Recommendations), Presentation	Hatice Kavçakar
Arena Simulation (Technical Modeling & Troubleshooting, Advanced Logic Design, System Bottleneck Analysis, What-IF Analysis & Improvement Scenarios, Corporate Communications for Temporary Full-Access Licensing)	Yaman Işıklı
Report (Event Flowcharts, Input Modeling, Distribution Fitting Results, Time Study, Scenario Analysis, Experimental Design & Setup), Presentation	Elif Açıktur

4. System Description & Conceptual Model

4.1. System Description

The system represents the departure process of passengers at Regional Airport (SRA). Passengers arrive at the airport either by private car or by being dropped off near the terminal entrance. This determines whether passengers need to walk from the parking area to the terminal or enter the departure hall directly.

After entering the terminal, passengers are assigned attributes and follow different paths depending on their check-in status and baggage condition. Passengers who have completed online check-in proceed directly to the security checkpoint, while the remaining passengers go to the check-in counters. Check-in is separated into different service types, and passengers are routed accordingly.

After check-in, passengers with checked baggage continue to the baggage drop area, while passengers without baggage proceed directly to the security checkpoint.

The security process is modeled as a multi-step system. Passengers first go through ID control and then proceed to the divesting area, where they prepare their belongings for screening. Passengers and their trays are then processed through parallel screening activities, including metal detectors and X-ray machines. At different stages, passengers may either pass directly or be directed to additional checks depending on the screening results. Additional procedures such as body search or manual bag inspection may be required. In some cases, passengers may be required to return to the check-in area due to issues detected during identity or security checks. After completing the screening process, passengers collect their belongings in the revesting area.

After leaving the security area, passengers are checked against their flight time. Those who are not on time are considered to have missed their flights and exit the system. Passengers who are on time continue through the connecting concourse and are directed to their assigned gates before walking to the gate area.

Finally, passengers reach their gates, which represents the end of the system. Boarding itself is not modeled as a detailed service process; instead, the model focuses on whether passengers are able to complete all pre-boarding steps and arrive at their gates on time.

4.2. Assumptions & Simplifications

In order to build a manageable and realistic simulation model, several assumptions and simplifications were made based on the available data and modeling requirements. These assumptions are mainly used to keep the model practical while still reflecting the overall behavior of the system.

First, the flight schedule data was handled at the flight level rather than at the individual passenger level, and individual passengers were generated from total passenger counts using the provided arrival distributions. This allows the model to represent passenger arrivals over time while staying consistent with the dataset. Each passenger is treated as an independent entity, so no interaction effects between passengers are considered. Furthermore, passenger types were simplified into two categories. It was assumed that business-type passengers do not have checked baggage, and passengers without baggage were treated as business-type passengers. In other words, all passengers without baggage were classified as business-type passengers in the model. Although this may not fully reflect real-world behavior, it still helps separate different passenger flows more clearly.

Passenger walking speed was modeled using a triangular distribution with parameters (3, 4, 5) feet per second. This reflects the fact that passengers move at different speeds—some slower, some faster, and most around an average level. Since detailed data was not available, this distribution provides a reasonable approximation. Based on these speeds and the given distances, walking times to different areas in the terminal were calculated. Passengers who complete online check-in were assumed to skip both the check-in and baggage drop processes and go directly to the security checkpoint. This helps simplify the passenger flow and keeps the model easier to manage.

The “day of week” variable in the dataset was not used, as it was found to be inconsistent with actual calendar dates. Instead, the exact flight dates provided in the dataset were used when needed, and no distinction was made between weekdays and weekends.

All passengers were assumed to generally follow a forward process flow through the system. However, in some cases, passengers may be redirected to previous stages, such as returning to the check-in area due to issues during identity or document verification, or may undergo additional screening procedures when required.

All service resources were assumed to be continuously available throughout the simulation, meaning that equipment failures, breakdowns, or staff shortages were not considered. Queues were assumed to follow a first-in-first-out (FIFO) discipline, and their capacities were considered large enough so that no passenger is blocked from entering a queue.

Another important assumption is related to the gate closing rule. In the model, the gate is assumed to close 10 minutes before the scheduled flight departure time. Therefore, a passenger is considered on time only if the passenger reaches the gate control point at least 10 minutes before departure. Otherwise, the passenger is recorded as a missed flight.

Finally, boarding operations were not modeled as a detailed service process; instead, the model focuses on the departure process up to the point where passengers reach their assigned gates. In addition, although the space requirements of the proposed improvements were considered, possible building expansion or construction costs were not included in the analysis and were assumed to be outside the scope of this study.

4.3. Conceptual Model Components

The conceptual model describes how the passenger departure process operates and how passengers move through the system from arrival at the airport to either reaching their assigned departure gates or leaving the system after missing their flights. The system boundary starts with passenger arrival at the airport and ends when passengers reach their assigned gates or exit the airport due to not being on time.

Passengers enter the system according to the given arrival patterns and follow different paths depending on their check-in status and baggage condition. Passengers who complete online check-in proceed directly to the security checkpoint, while others first go through the check-in process. After check-in, passengers with baggage continue to the baggage drop area, whereas those without baggage move directly to security.

The security process is represented as a multi-step process rather than a single service block. Passengers first go through identity control and then move to the divesting area, where they prepare their belongings for screening. After that, passengers and their trays are processed through screening activities such as metal detectors and X-ray machines. Depending on the screening results, some passengers may go through additional procedures such as body search or manual bag inspection. In some cases, passengers may also be redirected to previous stages, such as returning to the check-in area due to issues during identity or security checks.

After completing the security process, passengers are checked based on their scheduled flight time. Passengers who are not on time are considered to have missed their flights and leave the system. Passengers who are on time continue through the connecting concourse and walk toward their assigned gates. Since arrival patterns, service times, and passenger routing decisions vary, the model represents a system where flow, waiting times, and resource usage change throughout the day.

4.3.1. Entities

The main entity in the system is the passenger. Passengers are generated based on flight schedules and arrival distributions, and each passenger is assigned attributes such as arrival time, baggage status, check-in type, flight time, and assigned gate. These attributes determine the route followed in the system, especially whether a passenger goes through check-in and baggage drop or proceeds directly to security. The flight time attribute is also used for the on-time control after security. Passengers are modeled as independent entities and move through the system individually based on their own characteristics.

4.3.2. Resources

The system includes check-in counters, baggage drop stations, and several security-related resources. Security resources include ID control stations, divesting areas, X-ray machines, metal detectors, body search stations, manual bag inspection stations, and revesting areas. Each resource serves passengers according to its capacity, and queues may form when demand exceeds available capacity. These resources are used to represent the main service points where delays and congestion may occur.

4.3.3. Activities

Passengers go through several main activities, including walking from the parking area if needed, check-in, baggage drop, security screening, walking through the connecting concourse, and walking to the assigned gate. The security activity includes multiple steps such as identity control, divesting, screening, possible additional checks, and revesting. Each activity requires time, and depending on resource availability, passengers may need to wait before receiving service.

4.3.4. Events

Key events in the system include passenger arrival, start and completion of check-in, completion of baggage drop, entry to security, completion of security screening, on-time control

after security, and exit from the system. A passenger may exit the system either by reaching the assigned gate on time or by leaving the airport after missing the flight. These events represent the main points where the state of the system changes over time.

4.3.5. Queues and System States

Queues form at check-in, baggage drop, and different security stages when passengers wait for available resources. Service is provided on a first-in-first-out (FIFO) basis. Queue lengths vary over time depending on passenger arrivals, service times, and available capacity.

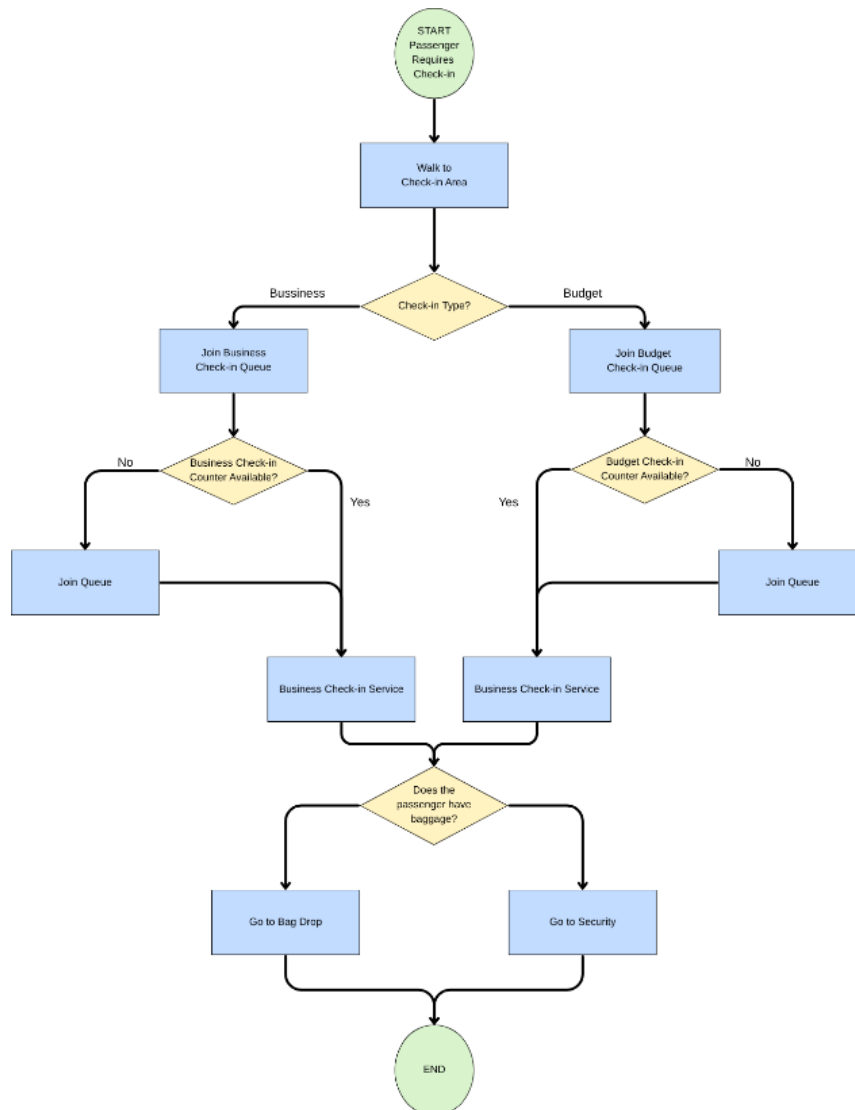
The system state can be described using measures such as the number of passengers waiting in each queue, the number of passengers being served, whether resources are busy or idle, and the number of passengers who miss their flights. These states are used to evaluate system performance, particularly in terms of waiting times, resource utilization, system load, and missed flights.

4.4. Event Flowcharts

To support the conceptual model, two representative event flowcharts were prepared for the check-in and revesting processes. These flowcharts show the main routing decisions, waiting points, service activities, and exit paths followed by passengers in the model.

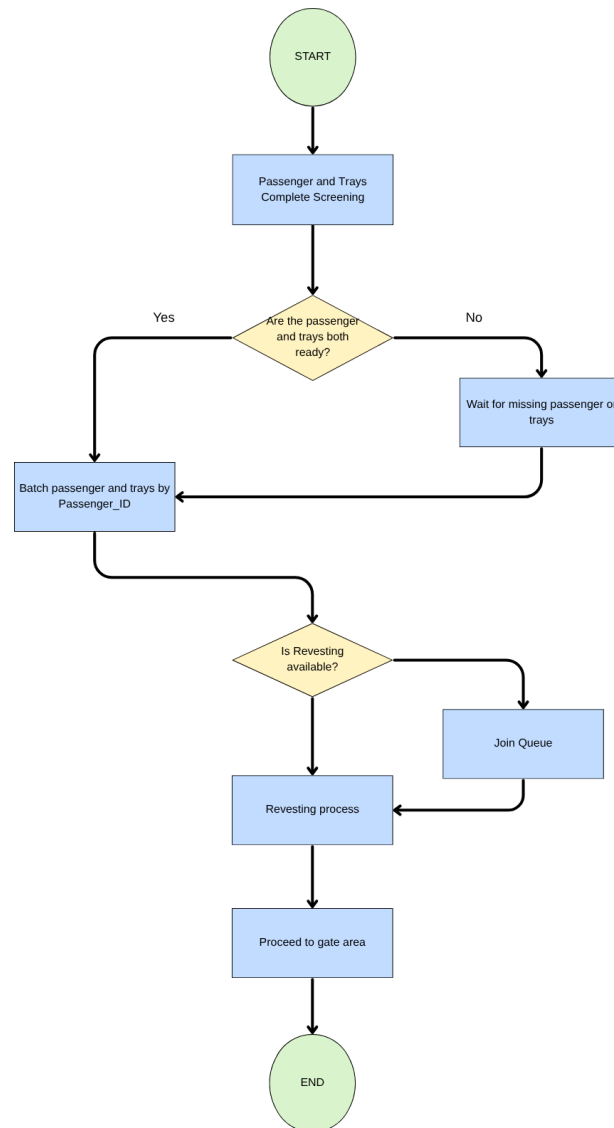
The first flowchart focuses on the check-in process. In this process, passengers who require check-in first move to the check-in area and are then separated according to their passenger type. Business and budget passengers follow different check-in paths, which allows the model to represent separate queues and service operations for these passenger groups. After completing the check-in service, passengers are routed based on whether they have baggage. Passengers with baggage continue to the bag drop process, while passengers without baggage proceed directly to the security area.

Figure 1. Event flowchart of the check-in process including passenger type routing, queueing, service, and baggage decision.



The second flowchart represents the revesting process after security screening. In this part of the system, passengers and trays may complete their screening activities at different times. Therefore, the model includes a synchronization step before the passenger can continue. If either the passenger or the trays are not ready, the entity waits until both are available. After synchronization, the passenger enters the revesting process. If the revesting resource is busy, the passenger waits in the revesting queue. Once revesting is completed, the passenger proceeds toward the gate area.

Figure 2. Event flowchart of the revesting process including passenger-tray synchronization, queueing, and continuation to the gate area.



Overall, these two flowcharts help explain the main logic behind the Arena implementation. The check-in flowchart shows how passengers are separated and routed before security, while the revesting flowchart shows how passenger-tray synchronization and queueing are handled after screening.

5. Input Modeling

The validity of a simulation model depends heavily on how well the input data reflects the stochastic behavior of the real system. In this study, the provided data sets were analyzed using the Arena Input Analyzer in order to determine suitable input models for the simulation. The

objective of this stage was to represent each process with an input model that is both statistically acceptable and practically realistic for implementation in Arena.

5.1. Verification of IID Properties

Before fitting probability distributions to the input data, the observations were examined with respect to the independence and identical distribution (IID) assumptions. This step is important because distribution fitting is more meaningful when the data do not exhibit strong serial dependence or structural instability. To illustrate this verification stage, the Revesting process was selected as a representative example.

To assess the independence assumption, the data were examined using a scatter plot and an autocorrelation plot. The scatter plot was used to visually inspect whether the observations exhibited any systematic trend, clustering, or repetitive pattern over the sequence of data. As shown in Figure 3, the Revesting observations are dispersed irregularly across the sequence without any clear upward or downward trend. This visual pattern suggests that the data do not contain an obvious structural dependency and supports the assumption of independence.

The autocorrelation plot provided additional evidence regarding serial dependence. As shown in Figure 4, the autocorrelation values for the examined lags remain within the 95% confidence limits. This indicates that no statistically significant serial correlation is present in the data. Therefore, the Revesting observations were considered sufficiently independent for input modeling purposes.

To assess the identical distribution assumption, the sample was divided into four subgroups, and the subgroup means and standard deviations were compared. The summary statistics are presented in Table 1. Since the subgroup averages and standard deviations are reasonably close to one another, the data were considered to be generated under a stable process. This supports the assumption that the observations were drawn from the same underlying distribution.

Overall, the graphical and numerical results do not provide strong evidence against the IID assumption for the representative Revesting data set. The scatter plot does not show a visible trend or clustering pattern, the autocorrelation values remain within the confidence limits, and the subgroup summary statistics are reasonably consistent across the sample. Accordingly, the IID assumption was considered acceptable, and the Revesting data were deemed suitable for the subsequent distribution-fitting stage.

Figure 3. Scatter plot of the Revesting process times used to visually assess the independence assumption.

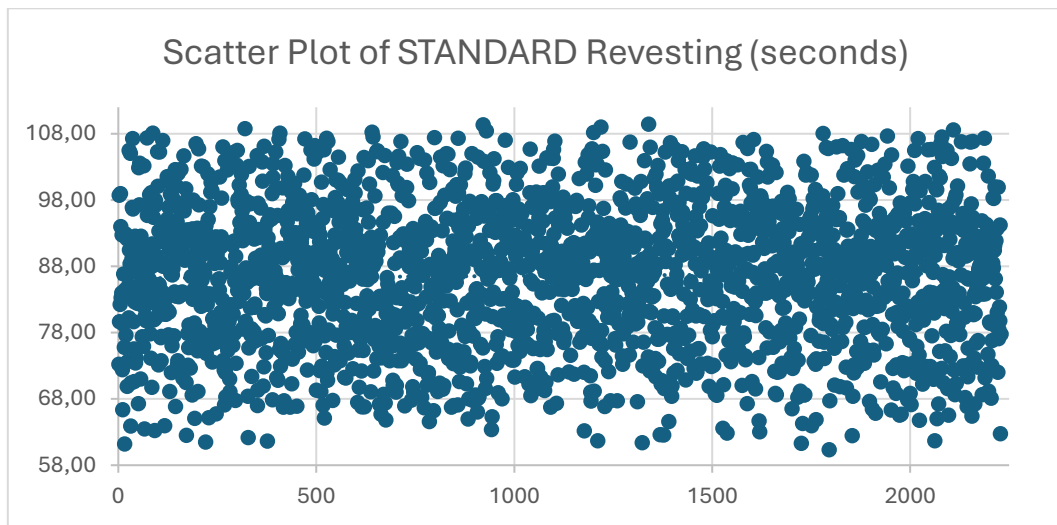


Figure 4. Autocorrelation plot of the Revesting process times used to assess serial dependence.

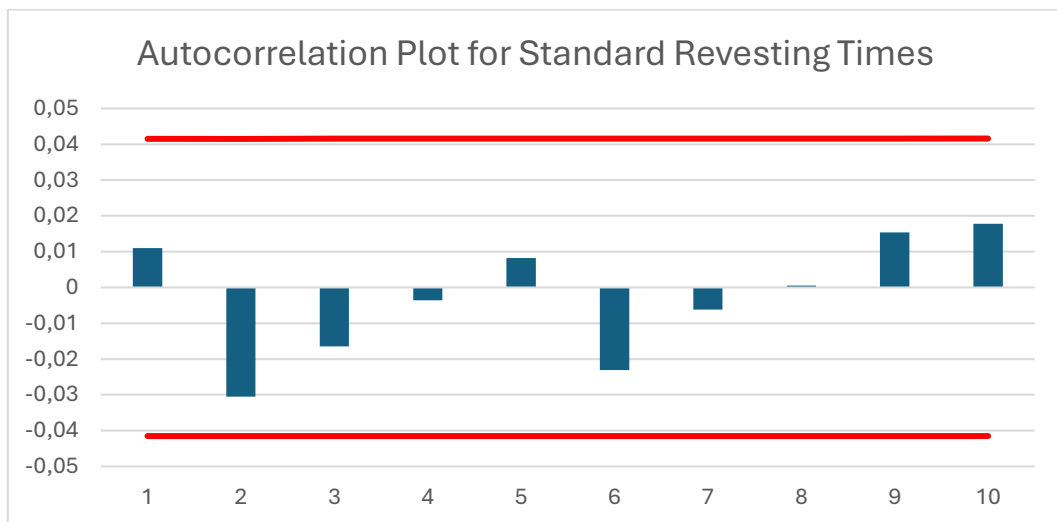


Table 1. Subgroup summary statistics used to assess the identical distribution assumption for the Revesting data.

Subgroup	Mean	Standard Deviation
GROUP 1	86.52	10.56573
GROUP 2	85.98	10.47521
GROUP 3	87.32	10.11982
GROUP 4	86.52	10.30618

5.2. Summary of Distribution Fitting Results

All input processes were analyzed using the Arena Input Analyzer to determine suitable input models for the simulation. For each data set, the Fit All option was used to examine candidate theoretical distributions and compare their statistical performance. For processes that displayed

a regular shape and acceptable goodness-of-fit results, the final selection was based on a combination of square error, Chi-Square goodness-of-fit results, Kolmogorov–Smirnov (K–S) test results, and visual agreement between the fitted curve and the histogram.

For several security-related processes with moderate sample sizes and relatively stable shapes, theoretical distributions such as Triangular and Weibull provided satisfactory results and were directly adopted in the Arena model. In these cases, both the statistical test results and the visual fit supported the selected distributions. However, not all data sets were equally suitable for a simple theoretical representation. In particular, some larger and more irregular data sets, especially those related to Check-In and Bag Drop, showed greater sensitivity in formal goodness-of-fit testing. Although certain theoretical distributions still produced relatively low square error values and visually plausible fits, they did not always preserve the observed structure of the original data adequately.

For this reason, the final modeling decision was not based solely on a single goodness-of-fit statistic. Instead, statistical fit, visual consistency, and practical representativeness for simulation were considered together. When a theoretical distribution was judged to be sufficiently accurate and interpretable, it was adopted directly. When the observed data pattern was more complex or irregular, an empirical representation was preferred in order to preserve the original behavior of the sample more faithfully in the Arena model.

Table 2. Summary of the distribution fitting results for the input processes used in the simulation model.

Input Process	Selected Distribution	Arena Expression	Square Error	Fit Decision
ID Check	Triangular	TRIA(5, 9.78, 20)	0.000301	Strong Fit ($p > 0.05$)
Revesting	Triangular	TRIA(60, 90, 110)	0.000305	Strong Fit ($p > 0.05$)
Manual Bag Scan	Weibull	15 + WEIB(23.1, 2.42)	0.001855	Strong Fit ($p > 0.05$)
Divesting	Triangular	TRIA(10, 29.7, 40)	0.000293	Strong Fit ($p > 0.05$)
Metal Detector	Triangular	TRIA(5, 10.1, 12)	0.000339	Strong Fit ($p > 0.05$)
Body Search	Weibull	35 + WEIB(5.7, 2.79)	0.002828	Strong Fit ($p > 0.05$)

Check-In (Budget)	Empirical	CONT(0.000, 1.000, 0.000, 1.225, ...)	-	Empirical Continuous Data
Check-In (Business)	Empirical	CONT(0.000, 0.170, 0.000, 0.316, ...)	-	Empirical Continuous Data
Bag Drop	Empirical	CONT(0.000, 0.070, 0.000, 0.129, ...)	-	Empirical Continuous Data
Trays Needed	Empirical	DISC(0.202, 1, 0.601, 2, ...)	-	Empirical Discrete Data

As shown in Table 2, several security-related processes, such as Revesting, Manual Bag Scan, Divesting, Metal Detector, and Body Search, were represented adequately by standard theoretical distributions. These inputs were therefore modeled using the fitted Arena expressions shown in the table. However, the results also indicate that not all processes were equally suitable for a simple theoretical representation.

Figure 5. Arena Input Analyzer output for the Standard ID Check process fitted with a triangular distribution.

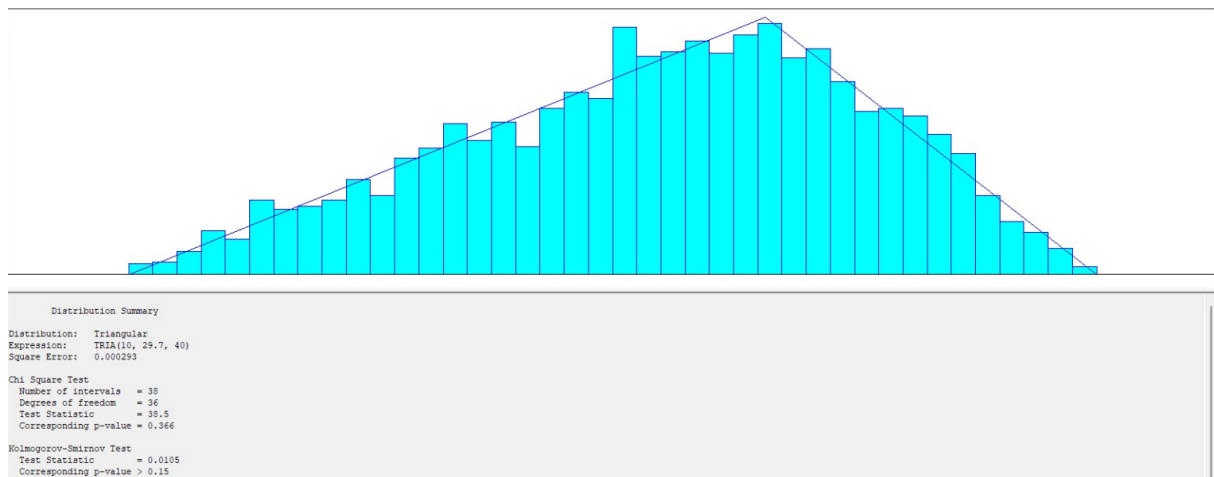


Figure 6. Arena Input Analyzer output for the Budget Check-In process, illustrating the limitations of a theoretical fit and supporting the use of an empirical distribution.

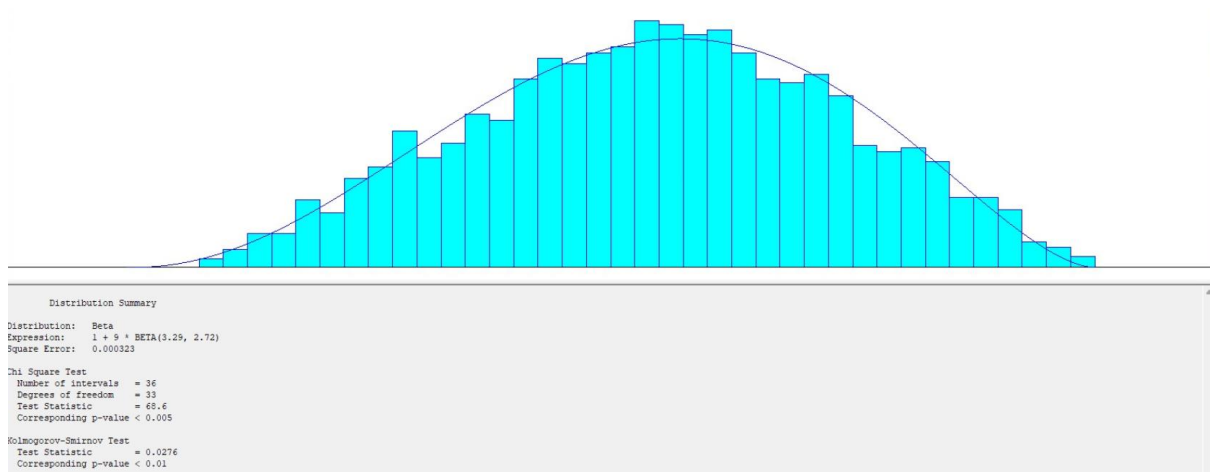


Figure 5 and Figure 6 illustrate the reasoning behind this mixed modeling approach. In Figure 5, the Standard ID Check data show a strong visual agreement with the fitted triangular distribution, and the associated test results also support the suitability of that model. In Figure 6, however, the Budget Check-In data exhibit a more irregular shape. Although a theoretical distribution may still produce a relatively low square error, the formal goodness-of-fit results are less supportive. In such cases, using an empirical distribution allows the simulation model to preserve the original behavior of the observed data more accurately.

Accordingly, processes with strong theoretical fits were modeled using standard probability distributions, while processes with more irregular or highly sensitive large-sample structures were represented using Arena's empirical expressions, such as `CONT(...)` and `DISC(...)`. This approach provided a more balanced representation of the system by preserving both statistical credibility and practical realism in the simulation model.

Note on large data sets: For processes such as Check-In and Bag Drop, the sample sizes were very large. In such cases, the Chi-Square and K-S tests become highly sensitive, and even relatively small differences between the observed data and the fitted theoretical curve may lead to low p-values. Therefore, these inputs were evaluated not only by formal test results but also by their square error values and the visual agreement between the fitted distribution and the histogram.

Overall, the input modeling stage was designed to balance statistical fit and practical realism. Theoretical distributions were adopted when they provided both acceptable goodness-of-fit results and a visually consistent representation of the observed data. When this was not the case, empirical Arena expressions were preferred to preserve the original data structure more

faithfully. To further support this approach, the next subsection presents an explicit Chi-Square goodness-of-fit test for one selected input process.

5.3. Explicit Chi-Square Goodness-of-Fit Test for the Standard Metal Detector

To provide an explicit statistical example of the input modeling procedure, a Chi-Square goodness-of-fit test was carried out for the Standard Metal Detector processing times. According to the distribution fitting stage, the Standard Metal Detector data were best represented by a Triangular distribution. This subsection presents the manual Chi-Square test used to evaluate whether the selected distribution is statistically consistent with the observed sample.

The purpose of this test is to compare the observed class frequencies obtained from the sample with the expected frequencies generated under the fitted theoretical distribution. If the discrepancy between observed and expected values is not statistically significant, then the selected distribution can be accepted as a suitable representation of the input data for simulation purposes.

Hypotheses

The hypotheses for the test were defined as follows:

- **Null hypothesis (H_0):** The Standard Metal Detector processing times follow a triangular distribution, $TRIA(5, 10.1, 12)$.
- **Alternative hypothesis (H_1):** The Standard Metal Detector processing times do not follow the triangular distribution, $TRIA(5, 10.1, 12)$.

Test Setup

The sample consisted of 2229 observations. The fitted triangular distribution had a lower limit of 5 seconds, a mode of 10.1 seconds, and an upper limit of 12 seconds. The data were divided into 50 probability-based classes, and the expected frequency for each class was calculated as 44.58. The significance level was selected as 0.05.

Table 3. Main test parameters used in the Chi-Square analysis.

Parameter	Value
Sample size (n)	2229

Number of classes (k)	50
Lower limit (a)	5
Mode (m)	10.1
Upper limit (c)	12
Expected frequency per class	44.58
Significance level (α)	0.05

The class boundaries were obtained from the cumulative probabilities of the fitted triangular distribution. For each class, the observed and expected frequencies were compared, and the Chi-Square components were summed to obtain the final test statistic. Excerpts from the Excel-based class-by-class calculation table are provided in Appendix A. In particular, Figure A 2 presents a portion of the detailed calculation sheet used in the Chi-Square analysis.

Test Results

The resulting Chi-Square statistic was calculated as:

$$\chi^2 = 48.2768$$

with 46 degrees of freedom, producing a p-value of 0.3810.

Table 4. The summary results of the Chi-Square goodness-of-fit test for the Standard Metal Detector processing times.

Statistic	Value
Chi-Square statistic	48.2768
Degrees of freedom	46
p-value	0.3810
Decision	Fail to reject H_0

Statistical Decision and Interpretation

Since the p-value (0.3810) is greater than the selected significance level (0.05), the null hypothesis cannot be rejected. Therefore, there is no statistically significant evidence to conclude that the Standard Metal Detector processing times deviate from the proposed triangular distribution.

At the 95% confidence level, the triangular distribution $TRIA(5,10.1,12)$ was considered an acceptable representation of the Standard Metal Detector processing times. Accordingly, this distribution was used in the Arena simulation model.

It should be noted that this test evaluates only the distributional fit of the data. The independence and identical distribution properties of the sample were examined separately in Section 5.1 through scatter plot and autocorrelation analysis.

6. Arena Model Design and Logic

The Arena simulation model was developed to represent the passenger departure process at Regional Airport (SRA). The model follows a discrete-event simulation structure, where each passenger is represented as an individual entity moving through the terminal system. The main purpose of the model is to evaluate how different operational stages such as arrival, check-in, baggage drop, security screening, and gate access affect passenger flow and the probability of missing a flight.

The model was designed to reflect the end-to-end passenger journey from airport arrival to the assigned boarding gate. Each passenger is assigned individual attributes such as arrival time, walking speed, passenger type, baggage status, flight departure time, and gate assignment. These attributes determine the path followed by each passenger in the system and allow the model to represent heterogeneous passenger behavior rather than treating all passengers identically.

The overall model logic is divided into four main parts:

1. Passenger arrival and attribute assignment
2. Check-in and baggage processing
3. Security screening process
4. Final time control and gate movement

This structure makes the model modular and easier to verify. It also allows scenario-based changes, such as increasing baggage drop capacity, adding kiosks, or using advanced security technology, to be tested without changing the entire model structure.

6.1. Passenger Arrival and Attribute Assignment

6.1.1. Preparation of Passenger Arrival Inputs from the Flight Schedule

Passenger arrivals were generated from the forecasted flight schedule instead of using a simple stationary interarrival distribution. This approach was preferred because passenger arrivals at an airport are strongly related to flight departure times. For this reason, the Excel file was used to convert the flight schedule into hourly passenger arrival inputs for the Arena model.

First, the cleaned flight schedule was prepared by combining the relevant flight records for the 2025, 2026, and 2027 planning years. Each flight record included information such as departure time, airline, terminal, gate group, and estimated passenger demand. After this step, summary sheets were created to examine the distribution of flights and passengers by year, month, hour, terminal, and gate group.

Then, an arrival probability matrix was used to estimate when passengers arrive at the airport before their flight departure time. The matrix divides the pre-departure period into time windows such as 120–110 minutes, 110–100 minutes, and 100–90 minutes before departure. Each window has a probability value that represents the expected share of passengers arriving during that interval. The highest probabilities are assigned to the middle part of the pre-departure period, especially around 80–70 minutes before departure. This reflects the assumption that most passengers do not arrive extremely early or extremely late.

Table 5. Arrival probability matrix used to distribute passengers before scheduled flight departure.

Window Before Departure (min)	Probability
120-110	1.0%
110-100	7.0%
100-90	12.0%
90-80	16.0%
80-70	20.0%
70-60	10.0%
60-50	9.0%
50-40	8.0%
40-30	9.0%
30-20	5.0%
20-10	3.0%
10-0	0.0%

For each flight, the passenger demand was distributed across these pre-departure time windows according to the arrival probability matrix. The arrival time for each group was calculated by subtracting the midpoint of the related time window from the flight departure time. In this way, departure-based passenger demand was transformed into arrival-based passenger demand.

The calculation can be summarized as follows:

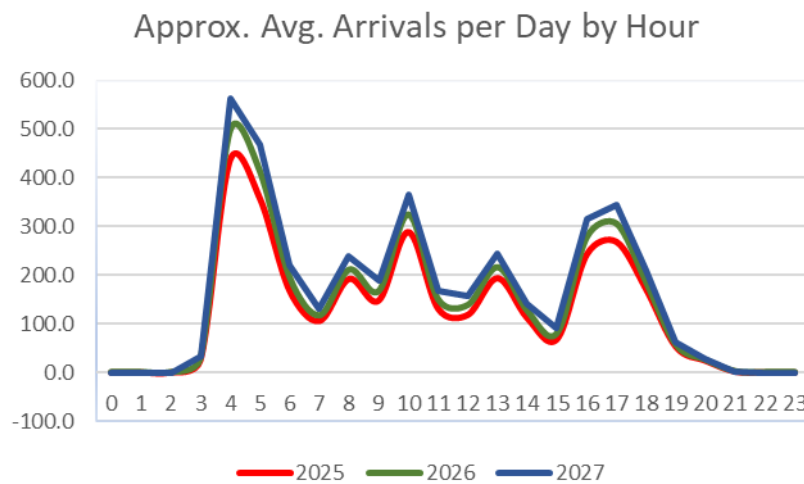
$$A_{f,w} = P_f \times p_w$$

where $A_{f,w}$ is the number of passengers assigned to time window w for flight f , P_f is the estimated passenger demand of flight f , and p_w is the probability of arriving in time window w .

After this calculation, the estimated passengers were aggregated by hour to obtain the average daily passenger arrival profile. This hourly arrival profile was then transferred to Arena as a Schedule and connected to the Create module. In this way, the Create module generated passengers according to the changing arrival intensity during the day instead of using a fixed interarrival time.

This schedule-based arrival structure allowed the model to represent time-dependent airport demand more realistically. Passenger arrivals increased during periods with higher flight-related demand and decreased during off-peak periods. Therefore, the model reflects the relationship between the flight schedule and passenger arrival behavior.

Figure 7. Average daily passenger arrivals by hour based on the forecasted flight schedule.



6.1.2. Passenger Creation and Attribute Assignment in Arena

The model begins with the arrival of passengers into the airport system. Passenger entities are generated using the Create module based on the arrival schedule prepared from the flight schedule and arrival probability matrix. After entering the model, each passenger is recorded using a Record module so that the total number of passengers can be tracked.

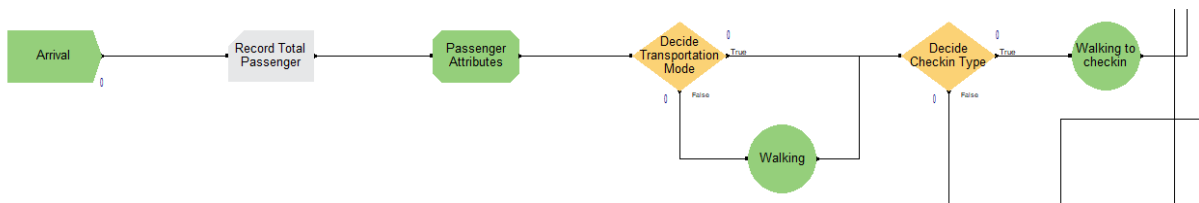
Immediately after arrival, the Assign module is used to define the main passenger attributes. These attributes are important because the model depends on passenger-specific characteristics rather than a single fixed process path. For example, one passenger may arrive by private car and need additional walking time, while another passenger may be dropped off closer to the terminal entrance. Similarly, some passengers may need manual check-in or baggage drop, while others may proceed directly toward security.

The assigned attributes include variables such as passenger type, flight time, gate group, baggage requirement, check-in requirement, transportation mode, and walking speed. These attributes determine the path that each passenger follows in the Arena model.

The transportation mode is represented using a Decide module. This decision determines whether the passenger enters the terminal directly or requires an additional walking process before reaching the next stage. Walking time is modeled separately so that terminal movement can affect the total passenger time in the system.

After the transportation mode is determined, the passenger is routed to the check-in decision logic. This stage decides whether the passenger can continue directly toward security or must first complete check-in-related processes.

Figure 8. Passenger arrival, attribute assignment, transportation mode decision, and initial check-in routing logic.



6.2. Check-In and Passenger Type Logic

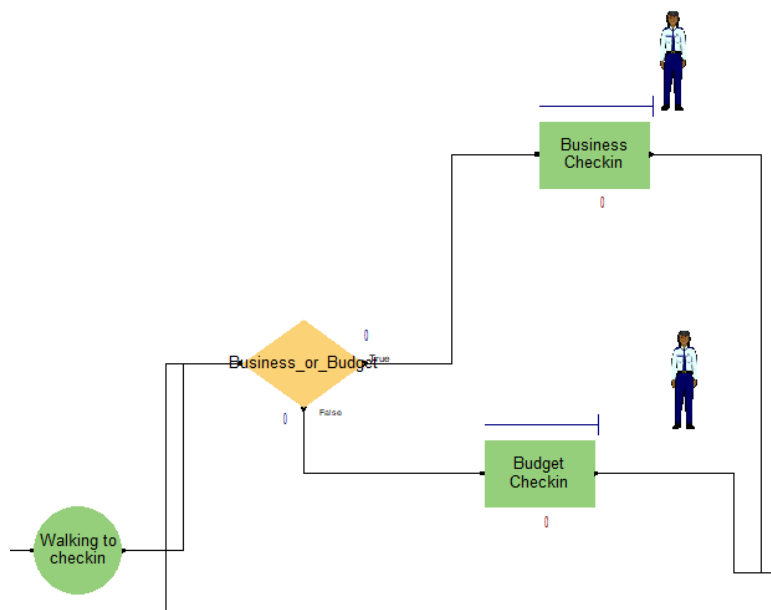
The check-in section of the model separates passengers according to their check-in requirements and passenger type. After the initial passenger attributes are assigned, the

Business_or_Budget Decide module is used to determine whether a passenger follows the business check-in path or the budget check-in path. This decision is based on the percentage values defined in the Assign module during passenger initialization.

Passengers who require check-in are routed to either the Business Checkin process or the Budget Checkin process. Both processes are modeled using Process modules with a Seize-Delay-Release structure. In this logic, a passenger first seizes an available check-in counter, spends a service time at the counter, and then releases the resource for the next passenger.

The check-in counters were divided between business and budget passengers in order to represent two different passenger groups separately. This structure allows the model to observe queue formation and waiting times for each group independently. It also helps identify whether the business or budget check-in flow creates a stronger bottleneck in the overall system.

Figure 9. Business and budget check-in separation in the Arena model.



After completing check-in, passengers continue to the baggage-related logic. Passengers with baggage are directed toward the baggage drop process, while passengers without baggage proceed toward the security checkpoint.

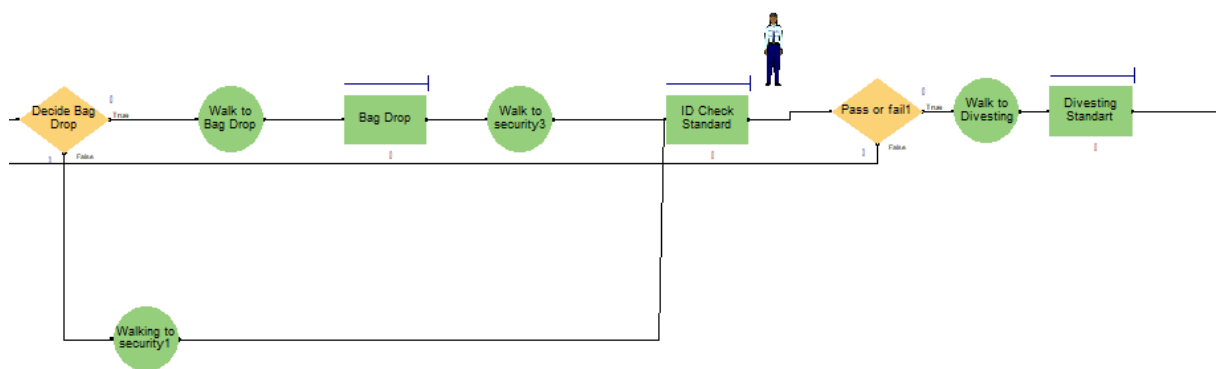
6.3. Baggage Drop Processing Logic

The baggage-related part of the model represents the processing of passengers who need to drop off checked baggage before entering the security area. This stage is important because baggage drop can create an additional queue before security and may become a bottleneck when passenger demand is high.

Passengers requiring baggage service are routed to the baggage drop process. This process is modeled using a Process module, where the passenger seizes the baggage drop resource, completes the baggage service time, and then releases the resource. After this stage, passengers continue toward security screening.

The baggage process was modeled separately from check-in because baggage handling and check-in do not always create the same operational load. By separating these stages, the model can identify whether delays are caused mainly by check-in counters, baggage drop capacity, or downstream security processes.

Figure 10. Baggage drop, ID check, and divesting logic in the Arena model.



6.4. Security Screening Logic

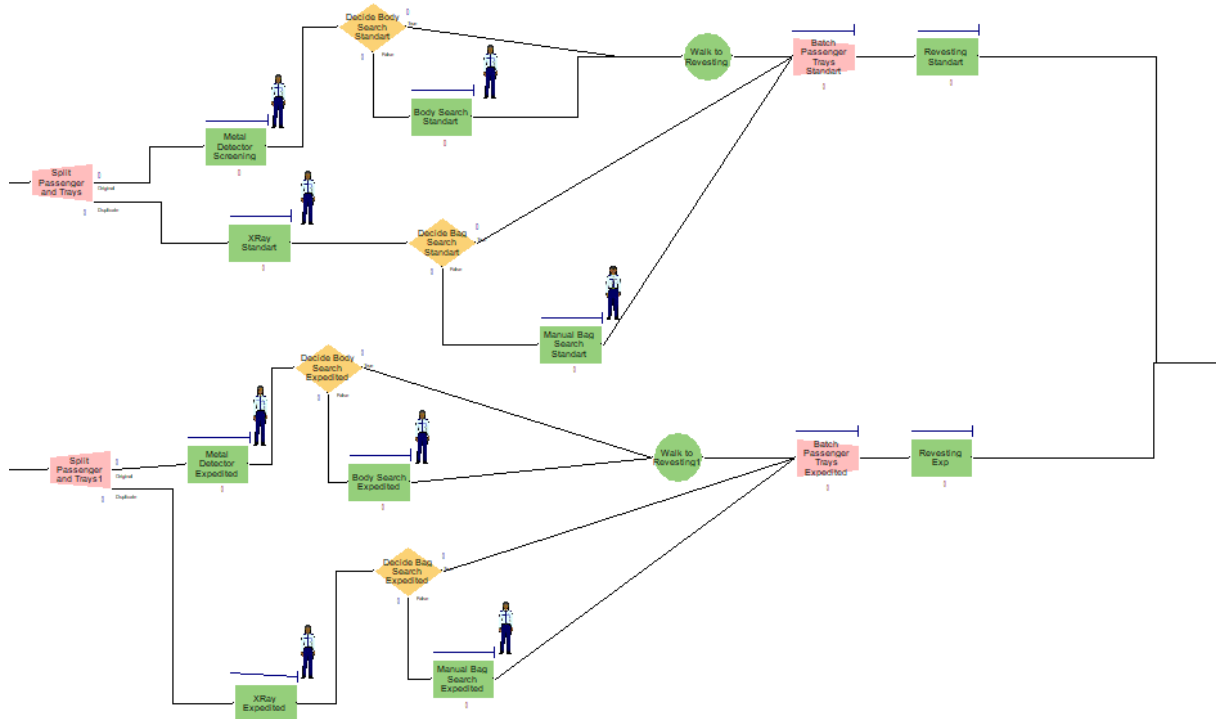
The security screening section is one of the most detailed parts of the Arena model. It was modeled as a multi-step process instead of a single service delay because real airport security consists of several connected activities. These include ID control, divesting, passenger screening, bag screening, possible secondary checks, and retesting.

In the model, passengers are separated into standard and expedited security flows. This distinction allows the simulation to represent different service channels and different security performance levels. Both flows follow a similar structure, but they are modeled with separate resources and process modules.

Before screening, the passenger and trays are separated using a Separate module. This is necessary because the passenger and the trays follow related but not identical processes. The passenger is processed through the metal detector or body scanner, while the trays are processed through the X-ray scanner. After these parallel processes are completed, the passenger and trays are brought together again using a Batch module before the retesting stage.

This approach is important because it prevents the model from unrealistically treating passenger screening and tray screening as a single activity. In reality, a passenger may wait for trays to finish screening, or trays may wait while the passenger completes additional checks. Therefore, the use of Separate and Batch modules improves the realism of the security process.

Figure 11. Standard and expedited security screening logic including passenger-tray separation, screening, additional checks, batching, and retesting.



The security process includes the following main activities:

- ID control
- Divesting
- Separation of passenger and trays
- Passenger screening through metal detector or body scanner
- Tray screening through X-ray scanner
- Additional body search if required
- Manual bag search if required
- Rejoining passenger and trays
- Revesting

The Decide modules after screening represent the possibility of additional inspection. If a passenger fails or is selected for extra screening, the passenger may be routed to body search.

Similarly, if a bag requires additional inspection, the tray may be routed to manual bag search. After these checks are completed, both flows continue toward the batching point.

6.5. Passenger-Tray Separation and Synchronization

In the security screening section, passengers and their trays were modeled as separate but related entities. After the divesting process, the Separate module was used to divide the passenger flow and tray flow. Passengers continued through the metal detector and possible body search, while trays continued through the X-ray scanner and possible manual bag search.

After these parallel screening activities were completed, the Batch module was used to bring the passenger and the related trays back together before the revesting stage. This was necessary because passengers cannot continue to revesting until both their personal screening and tray screening are completed.

The Batch module was configured using the *Passenger_ID* attribute, so each passenger was matched with their own trays. The batch size was defined as $1 + \text{Entity.Trays}$, where 1 represents the passenger and *Entity.Trays* represents the number of trays assigned to that passenger.

Figure 12. Batch module configuration used to synchronize each passenger with the associated trays based on *Passenger_ID*.

The screenshot shows a 'Batch' configuration window. The 'Name' field is 'h Passenger Trays Standart' and the 'Type' is 'Permanent'. The 'Batch Size' is '1 + Entity.Trays' and the 'Save Criterion' is 'First'. The 'Rule' is 'By Attribute' and the 'Attribute Name' is 'Passenger_ID'. The 'Representative Entity Type' is 'ePerson'. There is a 'Comment' field at the bottom, and 'OK', 'Cancel', and 'Help' buttons at the bottom right.

6.6. Terminal Movement, Time Control and Gate Routing

After completing the security process, passengers continue toward the gate area. At this stage, the model checks whether each passenger is still able to reach the flight on time. This control is represented using a Decide module. The condition used in the model is:

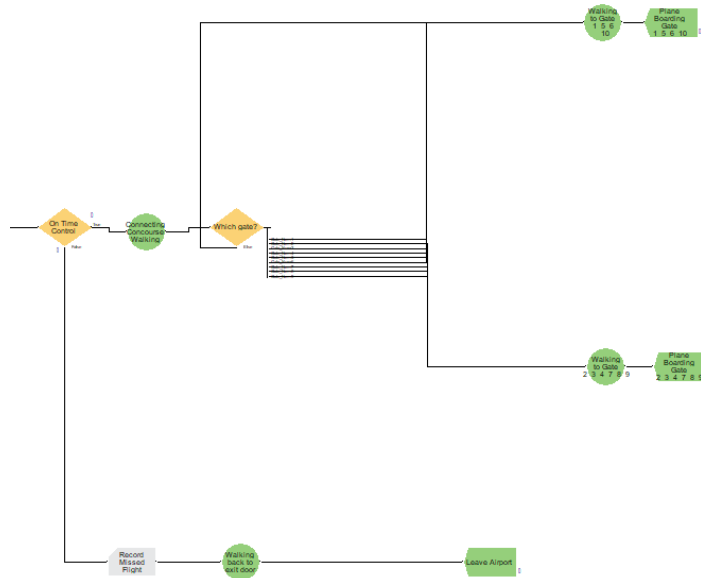
$$TNOW + 10 < Flight_Time$$

In this expression, $TNOW$ represents the current simulation time, and $Flight_Time$ represents the scheduled departure time of the passenger's assigned flight. The additional 10 minutes represents the gate closing buffer. Therefore, a passenger is considered on time only if they reach this control point at least 10 minutes before the flight departure time. If this condition is not satisfied, the passenger is counted as a missed flight and exits the system.

Passengers who are still on time continue through the terminal toward their assigned gates. The terminal movement logic includes walking through the connecting concourse and routing passengers to different gate groups. Walking time is modeled explicitly because travel distance inside the terminal affects the total time spent in the system. Each passenger has a $Walking_Speed$ attribute, assigned earlier in the model, and this attribute is used to calculate movement time. As a result, slower passengers may spend more time walking and have a higher risk of missing their flight.

The gate routing logic is represented using a Decide module. Based on the assigned gate number, passengers are directed to the appropriate gate group. After completing the final walking segment, passengers enter the gate area and then leave the system.

Figure 13. Terminal movement, on-time control, and gate routing logic in the Arena model.



6.7. Data Collection and Performance Measures

Record modules were used to collect the main performance measures of the model, especially the total number of passengers and the number of missed flights. This approach was preferred

instead of relying only on default Arena outputs, because the security section includes passenger-tray separation and synchronization.

After the terminal and security processes, passengers are checked with the on-time condition. If a passenger cannot reach the gate before the required time, the passenger is recorded as a missed flight. Otherwise, the passenger continues to the gate completion path.

Finally, Dispose modules remove passengers from the system after their relevant data have been recorded. This ensures that both successful completions and missed flights are counted consistently.

7. Verification & Validation

Verification and validation were conducted to evaluate the credibility of the Regional Airport (SRA) simulation model. Verification focuses on whether the model was implemented correctly in Arena, while validation focuses on whether the model represents the real airport system with reasonable accuracy. In this study, verification was performed through logical checks, animation tracing, module-level testing, and counter control. Validation was mainly based on input modeling checks and comparison of the base model output with the missed flight rate provided in the project description.

7.1. Verification of the Arena Model

The Arena model was verified by checking whether each part of the passenger flow behaved according to the conceptual model. The model was first examined module by module, including passenger arrival, transportation mode decision, check-in routing, baggage drop, security screening, terminal walking, gate selection, and final system exit. Arena animation was used to observe whether passengers followed the correct paths and whether queues formed at the expected locations.

Several Decide modules were checked to confirm that routing decisions were working correctly. For example, passengers were separated according to their transportation mode, check-in type, passenger type, baggage requirement, security lane type, additional search requirement, and assigned gate. The *Business_or_Budget* decision was also verified to ensure that passengers were correctly routed to the appropriate check-in process based on the attributes assigned earlier in the model.

The check-in, baggage drop, and security processes were verified by observing the Seize-Delay-Release logic in the Process modules. This confirmed that passengers seized the correct resource, spent the assigned service time in the process, and released the resource before moving to the next stage. Queues were also monitored to ensure that waiting passengers were held before service when resources were busy.

Special attention was given to the security section because the model includes passenger-tray separation. Separate modules were used to divide passengers and trays into different flows, and Batch modules were used to synchronize them again before revesting. This logic was verified through animation to make sure that passengers did not continue to the revesting stage before both the passenger screening and tray screening processes were completed.

The missed flight logic was also checked carefully. The model uses the current simulation time and the passenger's flight departure time to determine whether the passenger reaches the gate on time. Since the gate is assumed to close 10 minutes before departure, the passenger is considered on time only if the condition $TNOW + 10 \leq Flight_Time$ is satisfied. Otherwise, the passenger is counted as a missed flight and exits the system through the missed flight path.

Because the model includes split and batch logic, Arena's standard entity output statistics may not represent the actual number of passengers correctly. Therefore, Record modules were used to count the total number of passengers and the number of missed flights directly. This ensured that the main performance measure was calculated from the intended passenger-level counters instead of duplicated tray-related entities.

Overall, the verification process showed that the model logic was internally consistent. Passenger routing, resource usage, synchronization, walking-time calculations, gate selection, and missed flight counting all behaved according to the intended model structure.

7.2. Validation of the Model

Validation was performed to evaluate whether the simulation model represents the real system at an acceptable level. The input modeling stage provided the first part of validation. Before fitting distributions, the input data were examined for IID behavior using graphical and numerical checks. The scatter plot, autocorrelation plot, and subgroup statistics did not show strong evidence against the IID assumption. Therefore, the fitted input distributions were considered suitable for use in the Arena model.

The second part of validation was based on the base scenario output. The project description states that the airport normally experiences a missed flight rate in the range of 10% to 15%. Therefore, the base model was expected to produce a missed flight percentage within or close to this range under current operating conditions.

The base scenario was run for five independent replications. The results were calculated using the custom Record counters in the model. Table 6 presents the missed flight results for the base scenario.

Table 6. Base scenario missed flight results used for validation.

Replication	Missed Flights	Total Passengers	Missed Flight Rate
1	441	3552	12.42%
2	450	3535	12.73%
3	357	3484	10.25%
4	303	3453	8.77%
5	334	3494	9.56%
Total / Weighted Result	1885	17518	10.76%

The weighted missed flight rate was calculated as follows:

$$\text{Missed Flight Rate} = \frac{17518}{1885} \times 100 = 10.76\%$$

Figure 14. Cumulative total passenger count and missed flight count in the base scenario.

Total Passenger & Missed Flight Count

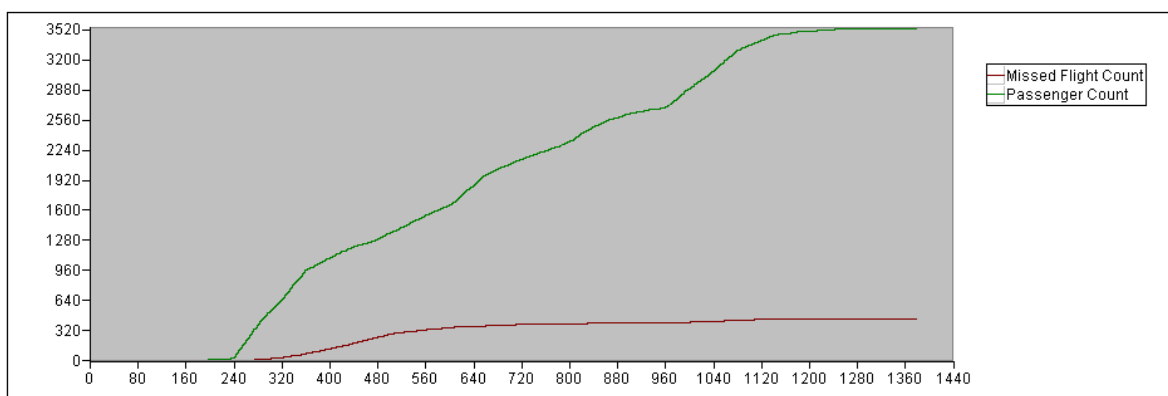


Figure 14 shows the cumulative passenger count and missed flight count during the base scenario run. The passenger count increases during the active passenger arrival period, while the missed flight count increases more slowly. This figure also supports the counter logic because it shows that the model separately tracks total passengers and missed passengers throughout the simulation.

The base model produced a missed flight rate of 10.76%. This value is within the 10%–15% range given in the project description. Therefore, the base model was considered valid enough to represent the current airport system and to be used as a reference point for improvement scenarios.

7.3. Replication-Based Reliability

Since the model contains random passenger arrivals, service times, passenger characteristics, and routing decisions, a single simulation run would not be sufficient to evaluate system performance. For this reason, the base model was run for five independent replications.

The replication results show some variation, which is expected in a stochastic simulation model. The lowest missed flight rate was 8.77%, and the highest missed flight rate was 12.73%. However, the weighted average across all replications was 10.76%, which is consistent with the expected missed flight range. This indicates that the model output is reasonable and stable enough for comparing alternative improvement scenarios.

As a result, the model was accepted as both verified and validated for the purposes of this study. The base scenario can therefore be used as the benchmark for evaluating future operational improvements such as check-in changes, baggage drop capacity increases, security technology upgrades, and terminal movement alternatives.

8. Experimentation and Scenarios

This section presents the experimental phase of the simulation study. After the base model was verified and validated, several improvement scenarios were tested to evaluate how different operational, spatial, and technological changes affect the missed flight rate at the Regional Airport (SRA). The main objective of this stage was to identify the most effective improvement strategy under the project's budget and space limitations.

8.1. Experimental Design and Setup

Before comparing the scenarios, the simulation settings were kept consistent across all experiments to ensure a fair comparison. The airport departure system was modeled as a terminating simulation because the operations are tied to a daily flight schedule. The system starts empty before the first passengers arrive and ends after the scheduled passenger flow is completed. Therefore, the initial empty condition is part of the real operating cycle, and no warm-up period was used.

The run length was selected to cover the full operating period represented by the flight schedule. Each scenario was run using 5 independent replications because the model includes stochastic elements such as passenger arrivals, service times, walking speeds, and routing decisions. The results were evaluated using the average and weighted average missed flight rates obtained from these replications.

The main performance measure used in the analysis was the missed flight rate, calculated as:

$$\text{Missed Flight Rate} = \frac{\text{MissedFlightCount}}{\text{TotalPassengerCount}} \times 100$$

Since the Arena model includes passenger-tray separation and duplication logic, Arena's default entity output statistics were not used as the main source of performance measurement. Instead, dedicated Record modules were used to count the total number of passengers and the number of passengers who missed their flights.

8.2. Scenario Evaluation and Results

The analysis started with the Base Case, which represents the existing airport system without any improvement. The base model produced a missed flight rate of 10.76%, which was consistent with the expected range given in the project description. This result was used as the benchmark for comparing all improvement scenarios.

Scenario 1 focused on improving the check-in stage by adding self-service kiosks and extra budget check-in capacity. This scenario reduced the missed flight rate to 5.54%, showing that the check-in area was one of the major bottlenecks in the original system. This result suggests that reducing early queue formation can significantly improve the overall passenger flow.

Figure 15. Budget check-in queue behavior in the base scenario.

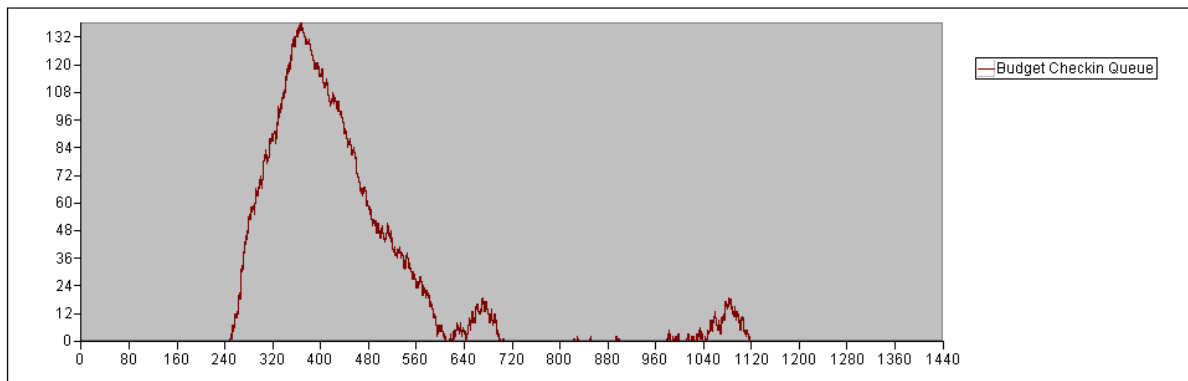


Figure 15 shows that the budget check-in queue becomes large during the active arrival period. This supports the interpretation that early-stage congestion is a major source of delay in the system.

Figure 16. Budget check-in queue behavior in Scenario 1 after the check-in improvement.

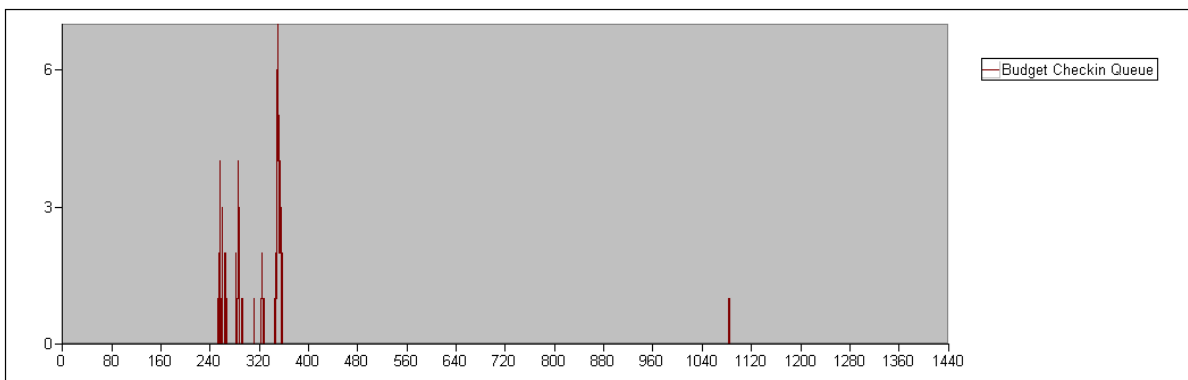


Figure 16 shows that the queue pressure at the budget check-in stage is reduced after the Scenario 1 improvement. This helps explain why Scenario 1 produced a strong reduction in the missed flight rate.

Scenario 2 tested whether adding more capacity to the standard security process would improve the system. However, the missed flight rate remained at 10.99%, which was slightly worse than the base case. This indicates that simply increasing manual security resources did not solve the main delay problem.

Scenario 3 improved the bag drop process by adding extra bag drop capacity. This reduced the missed flight rate to 9.27%. Although this was an improvement compared with the base case, the effect was limited. This suggests that bag drop was a relevant bottleneck, but not the only critical constraint in the system.

Scenario 4 tested the effect of moving sidewalks by reducing terminal walking time. The missed flight rate remained at 10.76%, which was almost identical to the base case. This result shows that passenger walking time was not the primary reason for missed flights. Instead, most delays were caused by service queues before passengers reached the gate area.

Scenario 5 evaluated the effect of introducing an advanced security lane into the system. In this scenario, the security process was improved by adding advanced screening technology and routing a large portion of passengers through the advanced lane. The missed flight rate decreased to 9.11%. Although this result was better than the base case, the improvement was still limited when compared with the high investment cost. Therefore, Scenario 5 showed that advanced security technology alone was not sufficient to be selected as the final solution.

Figure 17. Queue behavior of the standard, expedited, and advanced screening lanes in Scenario 5.

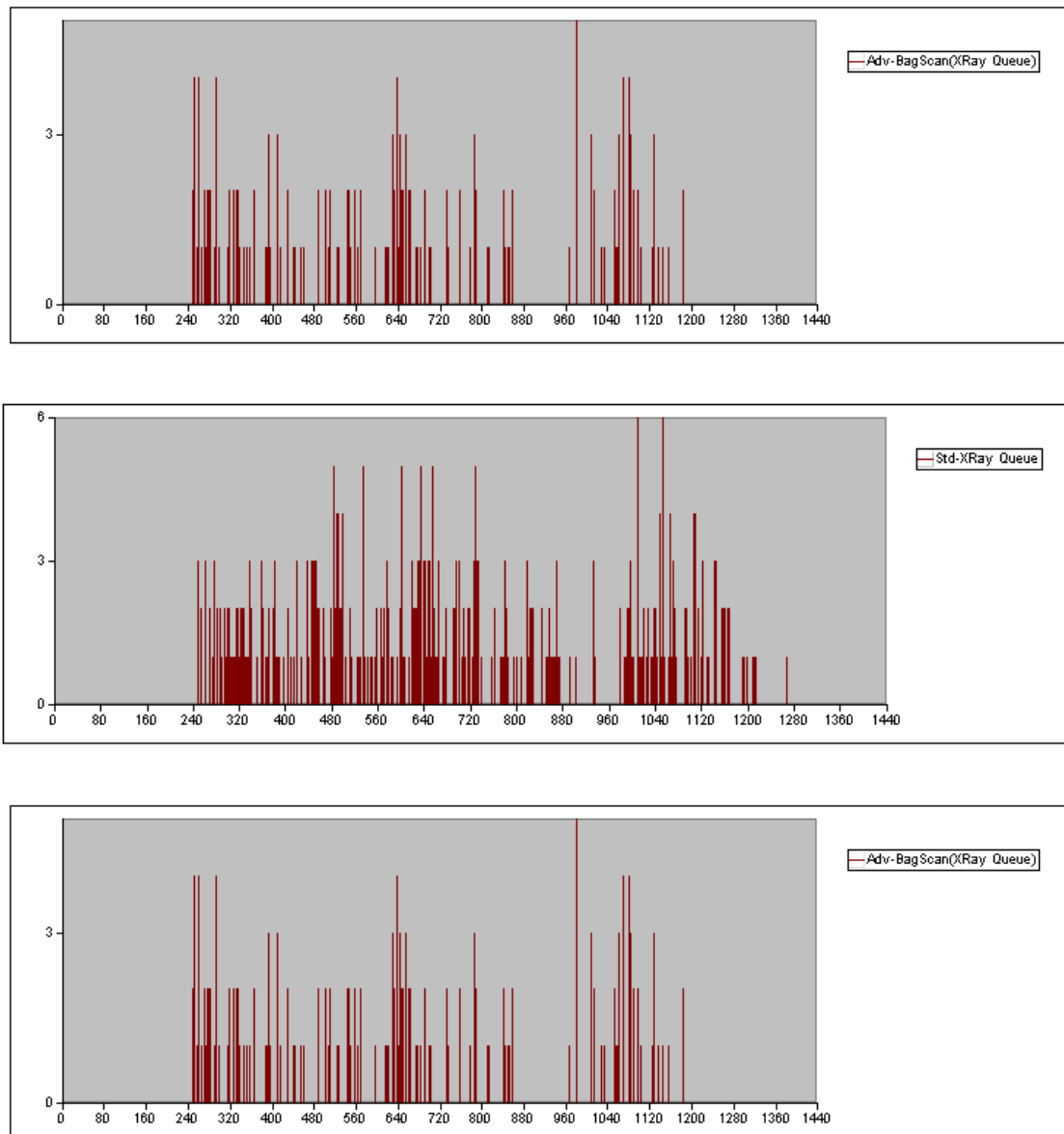


Figure 17 provides additional insight into the security performance of Scenario 5. The comparison shows how queue accumulation is distributed across the standard, expedited, and advanced screening lanes after the introduction of the advanced security technology. Although the advanced lane helps handle part of the screening workload, the remaining queues in the other security components indicate that the overall benefit is still limited at the system level. This helps explain why Scenario 5 improved the missed flight rate only moderately.

The best performance was obtained from the final integrated strategy, referred to as Scenario 6. This scenario combined multiple improvements instead of focusing on only one part of the system. The final version included 7 self-service kiosks, 1 additional bag drop scanner, 1 added advanced security lane, additional expedited X-ray capacity, and extra divesting and revesting capacity. This combined approach reduced the missed flight rate from 10.76% to 4.13%. The direct equipment cost was calculated as \$993,000, which keeps the scenario below the \$1,000,000 investment limit.

Table 7. Summary of scenario results and operational impact.

Scenario	Intervention Strategy	Estimated Cost	Missed Flight Rate	Operational Impact
Base Case	Current system configuration	\$0	10.76%	Baseline performance
Scenario 1	Check-in kiosks and additional budget counters	\$90,000	5.54%	Strong improvement
Scenario 2	Standard security capacity expansion	\$146,000	10.99%	Ineffective
Scenario 3	Bag drop capacity expansion	\$150,000	9.27%	Moderate improvement
Scenario 4	Moving sidewalks	\$140,000	10.76%	No meaningful improvement

Scenario 5	Single advanced security lane	\$863,000	9.11%	Limited improvement for high cost
Scenario 6	Integrated final strategy	\$993,000	4.13%	Best overall solution

8.3. Interpretation of Results

The scenario results show that the airport system contains shifting bottlenecks. When only one part of the system is improved, the delay often moves to another stage instead of disappearing completely. For example, improving the check-in process in Scenario 1 created a strong reduction in missed flights, but it did not fully solve the problem because passengers still faced downstream delays. Similarly, improving security or bag drop alone produced only limited benefits.

The moving sidewalk scenario showed that terminal walking distance was not the main cause of missed flights. Since the missed flight rate did not improve, the results suggest that the most critical delays occur before passengers reach the walking and gate assignment stage.

The final integrated strategy performed best because it improved multiple connected bottlenecks at the same time. By increasing self-service check-in capacity, improving bag drop capacity, and strengthening the security screening flow, Scenario 6 balanced the passenger flow across the system. This reduced the missed flight rate by approximately 61.6% compared with the base model.

Therefore, Scenario 6 was selected as the recommended improvement strategy. It provides the strongest operational improvement while remaining within the investment budget.

9. Results

The simulation results were evaluated by comparing the missed flight rate obtained from each scenario. The missed flight rate was selected as the main performance measure because the main objective of the study is to reduce the number of passengers who cannot reach their gates before the flight closing time.

The base model produced a missed flight rate of 10.76%, which was used as the reference value for evaluating the improvement scenarios. The numerical results are summarized in Table 8.

Table 8. Comparison of missed flight rates across all simulation scenarios.

Scenario	Missed Flight Rate	Difference from Base Case	Relative Change
Base Scenario	10.76%	0.00 percentage points	0.00%
Scenario 1	5.54%	-5.22 percentage points	-48.5%
Scenario 2	10.99%	+0.23 percentage points	+2.1%
Scenario 3	9.27%	-1.49 percentage points	-13.8%
Scenario 4	10.76%	0.00 percentage points	0.00%
Scenario 5	9.11%	-1.65 percentage points	-15.3%
Scenario 6	4.29%	-6.47 percentage points	-60.1%

As shown in Table 8, Scenario 6 produced the best overall performance, reducing the missed flight rate from 10.76% to 4.29%. This corresponds to a reduction of 6.47 percentage points, or approximately 60.1% compared with the base case. This result indicates that a combined improvement strategy is more effective than improving only one part of the airport process.

Among the individual improvement scenarios, Scenario 1 provided the strongest reduction, decreasing the missed flight rate to 5.54%. This shows that improvements in the early stages of the passenger flow, especially check-in-related operations, have a strong effect on the overall system performance.

Scenario 3 and Scenario 5 also improved the system, but their effects were more limited. Scenario 3 reduced the missed flight rate to 9.27%, while Scenario 5 reduced it to 9.11%. These results suggest that bag drop and advanced security improvements can support system performance, but they are not sufficient as standalone solutions.

Scenario 2 and Scenario 4 did not provide meaningful improvement. Scenario 2 resulted in a missed flight rate of 10.99%, which is slightly worse than the base case. Scenario 4 produced the same missed flight rate as the base case, 10.76%. These findings indicate that increasing standard security capacity or reducing walking time alone does not directly solve the main delay problem in the system.

To support the comparison of operational benefit and investment requirement, the cost-effectiveness of the scenarios was also considered. The cost per one percentage-point reduction in missed flight rate is summarized in Table 9.

Table 9. Cost-effectiveness comparison of the improvement scenarios.

Scenario	Estimated Cost	Reduction from Base Case	Cost per Percentage-Point Reduction
Scenario 1	\$90,000	5.22 percentage points	\$17,241
Scenario 3	\$150,000	1.49 percentage points	\$100,671
Scenario 5	\$863,000	1.65 percentage points	\$523,030
Scenario 6	\$1,083,000	6.47 percentage points	\$167,388

Table 9 shows that Scenario 1 is the most cost-effective individual improvement, since it provides a large reduction in missed flights with a relatively low investment cost. Although Scenario 6 produces the best operational result, it requires a much higher total investment. Therefore, the final decision depends on whether SRA prioritizes maximum performance improvement or cost-effective improvement.

Overall, the results show that missed flights are not caused by a single isolated process. Instead, delays are created by the interaction of multiple stages, including check-in, bag drop, and security. For this reason, the integrated improvement strategy in Scenario 6 provides the lowest missed flight rate. However, if the airport prefers a lower-cost improvement option, Scenario 1 appears to be the strongest alternative.

10. Analysis & Discussion

The results show that the missed flight rate is affected by several connected parts of the passenger departure process rather than by one isolated activity. As shown in Table 8, the base scenario produced a missed flight rate of 10.76%. This value was within the expected range

given in the project description, so it provided a reasonable baseline for comparing the improvement scenarios.

The strongest individual improvement was obtained in Scenario 1, where the missed flight rate decreased to 5.54%. This result suggests that delays occurring in the earlier stages of the system have a major effect on the final passenger outcome. When passengers lose time before reaching the security area, they have less remaining time to reach their gates. Therefore, improvements in check-in or early passenger processing can reduce downstream pressure and help more passengers reach their flights on time.

Figure 18. Budget check-in queue behavior in the base scenario.

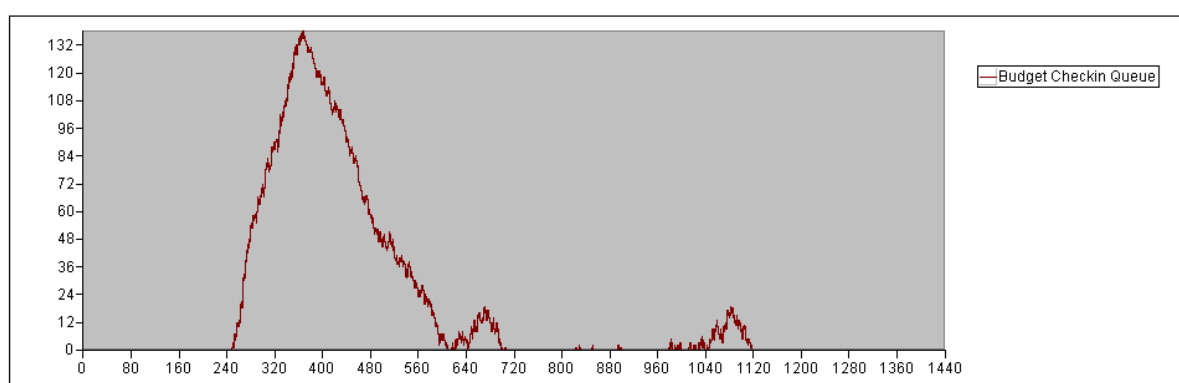


Figure 18 supports this interpretation by showing that the budget check-in queue can become very large during the active passenger arrival period. This indicates that early-stage congestion is an important source of delay in the system. Therefore, reducing check-in congestion can have a strong effect on the final missed flight rate.

By contrast, Scenario 2 did not improve the system. The missed flight rate increased slightly from 10.76% to 10.99%. This indicates that simply increasing or modifying one part of the standard security process does not necessarily improve the overall system. If the main source of delay is located before or after that point, then a local improvement may not create a visible system-level benefit. This result is important because it shows that operational decisions should be evaluated at the full-system level rather than focusing only on one station.

Scenario 3 and Scenario 5 provided moderate improvements. Scenario 3 reduced the missed flight rate to 9.27%, while Scenario 5 reduced it to 9.11%. These results show that baggage-related improvements and advanced security technology can support the system, but their individual effects are limited. In particular, Scenario 5 shows that advanced screening technology alone is not enough to solve the missed flight problem. Although it improves

passenger flow, the high investment cost makes its benefit less attractive when compared with lower-cost alternatives.

Scenario 4 produced the same missed flight rate as the base scenario, 10.76%. This suggests that reducing walking-related time or improving terminal movement alone does not address the main bottleneck of the system. Walking time still matters because passengers must reach their gates before the closing time, but the results suggest that most of the delay is created before passengers enter the final terminal movement stage.

The best overall result was obtained in Scenario 6, where the missed flight rate decreased to 4.29%. This scenario performed better because it combined multiple improvements instead of changing only one part of the system. The result supports the idea that the airport departure process is an interconnected system. A passenger may experience delay at check-in, bag drop, security, and walking stages. Therefore, a combined improvement strategy can reduce delay accumulation more effectively than a single improvement.

From a cost-effectiveness perspective, the results in Table 9 show that Scenario 1 is the most efficient individual alternative. It provides a large reduction in missed flights with a relatively low cost. Scenario 6 produces the lowest missed flight rate, but it also requires the highest investment. Therefore, the preferred solution depends on the management objective. If the goal is to achieve the maximum operational improvement, Scenario 6 is the strongest option. If the goal is to obtain a strong improvement with a more reasonable budget, Scenario 1 is more attractive.

Overall, the analysis indicates that SRA should not focus only on adding capacity to one station. The missed flight problem is mainly caused by the combined effect of several process delays. For this reason, the most successful solutions are the ones that reduce delay accumulation across the passenger journey. The simulation model provides a useful decision-support tool because it allows different improvement strategies to be tested before making real operational investments.

11. Conclusion & Recommendations

This project developed a discrete-event simulation model for the Regional Airport (SRA) departure system in Arena. The model represented the main passenger journey from arrival to gate, including transportation mode, check-in, bag drop, security screening, passenger-tray

synchronization, terminal walking, gate assignment, and missed flight recording. The main objective was to identify the causes of missed flights and evaluate possible improvement scenarios before making operational decisions in the real system.

The base model produced a missed flight rate of 10.76%, which is within the expected range stated in the project description. This result supports that the base model provides a reasonable representation of the current airport system. After the base model was verified and validated, several improvement scenarios were tested and compared according to their effects on the missed flight rate.

The results show that Scenario 1 is the most practical short-term recommendation. It reduced the missed flight rate to 5.54%, which is a major improvement compared with the base case. This indicates that improving the early passenger processing stages can significantly reduce delay accumulation before passengers reach the gate. Since delays at the beginning of the system affect all later stages, Scenario 1 provides a strong operational benefit without requiring the broadest system-level change.

The best overall performance was obtained in Scenario 6, where the missed flight rate decreased to 4.29%. This scenario achieved the lowest missed flight rate because it combined multiple improvements rather than changing only one part of the system. However, this option would require higher implementation effort and greater investment. Therefore, Scenario 6 should be considered as a long-term improvement plan if airport management aims to minimize missed flights as much as possible.

Scenario 5, which introduced an advanced security lane, reduced the missed flight rate to 9.11%. Although this result is better than the base case, the improvement was limited compared with the expected cost of advanced screening technology. For this reason, advanced security alone is not recommended as the final solution. It may support system performance when combined with other improvements, but it is not strong enough as a standalone strategy.

Overall, the simulation results suggest that the missed flight problem at SRA is not caused by a single station only. Instead, it is the result of delay accumulation across several stages of the passenger journey, especially before and during security processing. Therefore, the recommended approach is to first implement the most practical and effective improvement, represented by Scenario 1, and then consider Scenario 6 as a broader long-term strategy.

In conclusion, the Arena simulation model provides a useful decision-support tool for evaluating airport improvement alternatives without disrupting real operations. Based on the results, SRA should prioritize improvements that reduce early-stage passenger delays and improve flow continuity through the terminal. This approach can reduce missed flights, improve passenger experience, and support more reliable airport operations.

12. References

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13. Appendices

Appendix A. Chi-Square Calculation Sheets

Figure A 1. Excel-based class-by-class Chi-Square goodness-of-fit calculations for the Standard Metal Detector processing times.

Sample size(n)	2229
Number of Classes (k)	50
Lower Limit (a)	5
Mode (m)	10.1
Upper Limit (c)	12
Peak Probability	0.728571429
Expected Frequency per Class	44.58

Figure A 2. Partial view of the Excel-based class-by-class Chi-Square calculation table used for the Standard Metal Detector processing times.

Class	Cum. Prob. Lower	Cum. Prob. Upper	Lower Bound	Upper Bound	Observed Freq.	Expected Freq.	Chi-Square Component
1	0	0.02	5	5.844985207	49	44.58	0.438232391
2	0.02	0.04	5.844985207	6.19498954	48	44.58	0.262368775
3	0.04	0.06	6.19498954	6.46355731	55	44.58	2.435540601
4	0.06	0.08	6.46355731	6.689970414	50	44.58	0.658959175
5	0.08	0.1	6.689970414	6.889444363	44	44.58	0.007545985
6	0.1	0.12	6.889444363	7.069782597	41	44.58	0.287492149
7	0.12	0.14	7.069782597	7.235620719	49	44.58	0.438232391
8	0.14	0.16	7.235620719	7.389979079	47	44.58	0.131368327
9	0.16	0.18	7.389979079	7.534955621	56	44.58	2.925446389
10	0.18	0.2	7.534955621	7.672077843	38	44.58	0.971206819
11	0.2	0.22	7.672077843	7.802498885	41	44.58	0.287492149
12	0.22	0.24	7.802498885	7.92711462	40	44.58	0.470533872
13	0.24	0.26	7.92711462	8.046637491	36	44.58	1.651332436
14	0.26	0.28	8.046637491	8.161645141	44	44.58	0.007545985
15	0.28	0.3	8.161645141	8.272613634	46	44.58	0.045231045
16	0.3	0.32	8.272613634	8.379940828	57	44.58	3.460215343
17	0.32	0.34	8.379940828	8.48396326	47	44.58	0.131368327
18	0.34	0.36	8.48396326	8.584968619	48	44.58	0.262368775
19	0.36	0.38	8.584968619	8.683205126	29	44.58	5.444961866
20	0.38	0.4	8.683205126	8.778888726	32	44.58	3.549941678
21	0.4	0.42	8.778888726	8.872208672	50	44.58	0.658959175
22	0.42	0.44	8.872208672	8.963311932	36	44.58	1.651332436
23	0.44	0.46	8.963311932	9.052406692	50	44.58	0.658959175
24	0.46	0.48	9.052406692	9.139565195	58	44.58	4.039847465
25	0.48	0.5	9.139565195	9.224926035	46	44.58	0.045231045
26	0.5	0.52	9.224926035	9.308596059	41	44.58	0.287492149
27	0.52	0.54	9.308596059	9.39067193	51	44.58	0.924549125
28	0.54	0.56	9.39067193	9.471241438	48	44.58	0.262368775
29	0.56	0.58	9.471241438	9.550384599	35	44.58	2.058689996
30	0.58	0.6	9.550384599	9.628174586	41	44.58	0.287492149
31	0.6	0.62	9.628174586	9.704678522	45	44.58	0.003956931
32	0.62	0.64	9.704678522	9.779958159	41	44.58	0.287492149
33	0.64	0.66	9.779958159	9.854070457	48	44.58	0.262368775
34	0.66	0.68	9.854070457	9.927068094	41	44.58	0.287492149
35	0.68	0.7	9.927068094	9.9989999	34	44.58	2.510910722
36	0.7	0.72	9.9989999	10.06991124	35	44.58	2.058689996
37	0.72	0.74	10.06991124	10.14043016	50	44.58	0.658959175
38	0.74	0.76	10.14043016	10.21338309	43	44.58	0.055998205
39	0.76	0.78	10.21338309	10.28944453	42	44.58	0.149313594
40	0.78	0.8	10.28944453	10.36904936	55	44.58	2.435540601
41	0.8	0.82	10.36904936	10.45274437	49	44.58	0.438232391
42	0.82	0.84	10.45274437	10.5412334	45	44.58	0.003956931
43	0.84	0.86	10.5412334	10.63544879	35	44.58	2.058689996
44	0.86	0.88	10.63544879	10.73667106	38	44.58	0.971206819
45	0.88	0.9	10.73667106	10.84674374	43	44.58	0.055998205
46	0.9	0.92	10.84674374	10.96849624	49	44.58	0.438232391
47	0.92	0.94	10.96849624	11.10669154	48	44.58	0.262368775
48	0.94	0.96	11.10669154	11.2706167	46	44.58	0.045231045
49	0.96	0.98	11.2706167	11.48424812	41	44.58	0.287492149
50	0.98	1	11.48424812	12	48	44.58	0.262368775

Figure A 3. Final decision of the Excel-based Chi-Square goodness-of-fit test for the Standard Metal Detector processing times.

Chi-Square Statistic	48.27680574
df (Degrees of Freedom)	46
p-value	0.380997731
DECISION	Fail to Reject H0

Appendix B. Additional Input Data Calculations

Figure B 1. Bag drop requirement ratio derived from the input data.

BagDropProcessTime (minute)									
0.00	0	2766	Ratio	45.57%	This is ratio for passengers don't have bags				
1.28	Total	6070							
0.00									

Figure B 2. Airline type classification and budget passenger ratio used in the check-in routing logic.

AirlineType	CheckInProcessingTime (minutes)						
Budget	7.30						
Budget	6.85	Budget	3930			Budget Ratio	64.74%
Business	2.47	Business	6070				
Business	4.20	Total	10000				
Business	0.92						
Budget	6.73						

Figure B 3. Security-area processing time comparison and routing probability calculations for Standard and Expedited passengers.

	COUNT	RATIO	IDCheck	Divesting Mean	MetalDetector Mean	Bag Scan	Revesting	Body Search	Manual Bag Scan
Standard	2229	67.46%	11.63	26.60059219	8.996742934	19.57580978	86.58462988	40.04138889	35.55165414
Expedited	1075	32.54%	11.52	26.51853023	9.062883721	19.40807442	86.82145116	40.18782609	35.42702703
TOTAL	3304								
ABS			0.009708	0.003084967	0.007351637	0.008568502	0.002735142	0.003657146	0.003505522

The Standard and Expedited security areas show very similar average processing times across all major steps, with differences generally below 1%. Therefore, it is reasonable to model these process times using pooled distributions rather than separate ones. The main difference between the two areas appears to come from capacity and station configuration rather than service-time behavior.

[illegible]

Appendix C. Arena Input Analyzer

Figure C 1. Arena Input Analyzer result for Budget Check-in Time a large input data set, where an empirical distribution was selected due to poor theoretical distribution fit.

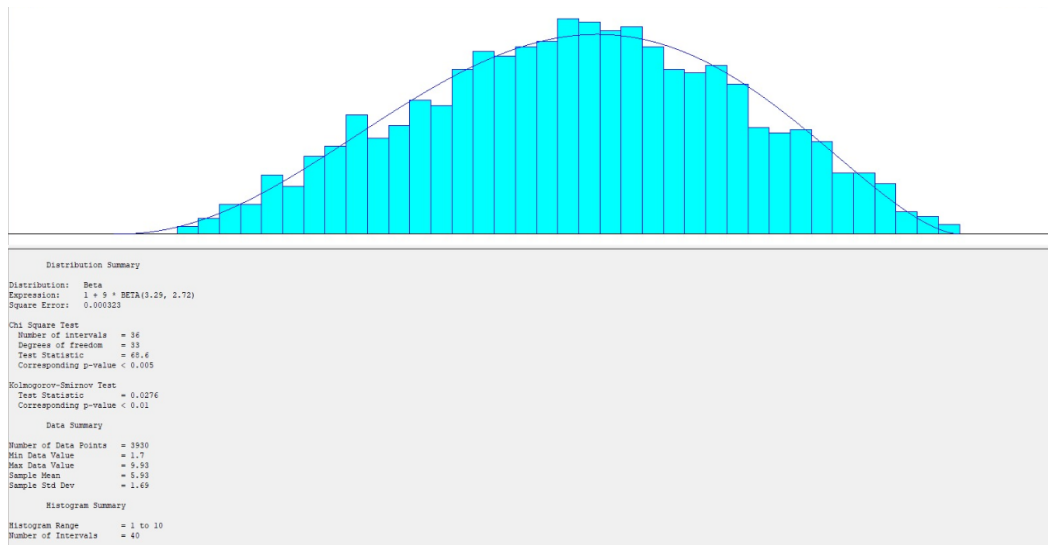


Figure C 2. Empirical distribution generated for the trays needed input process using Arena Input Analyzer.

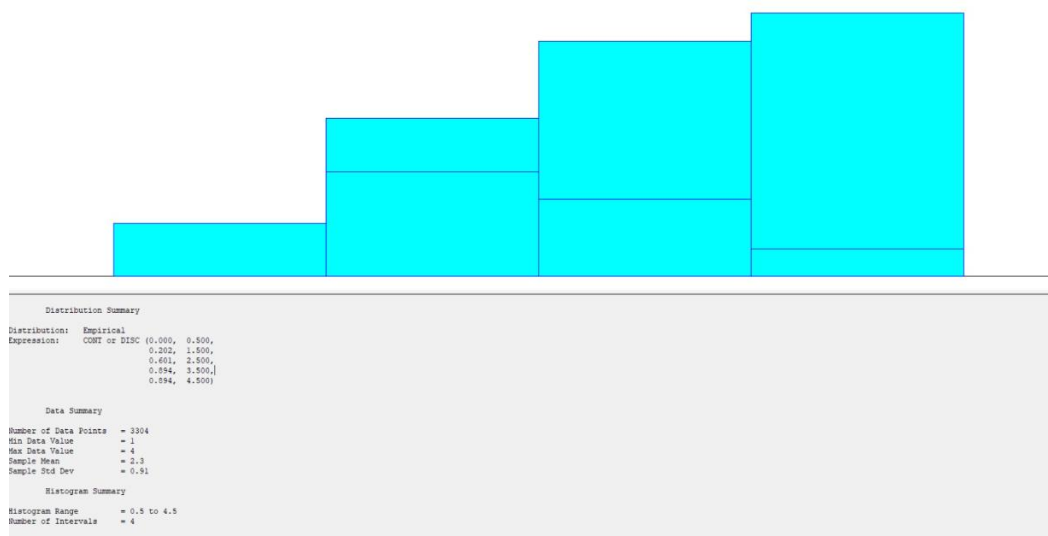


Figure C 3. Arena Input Analyzer result for the divesting process, where a triangular distribution was fitted to the observed data.

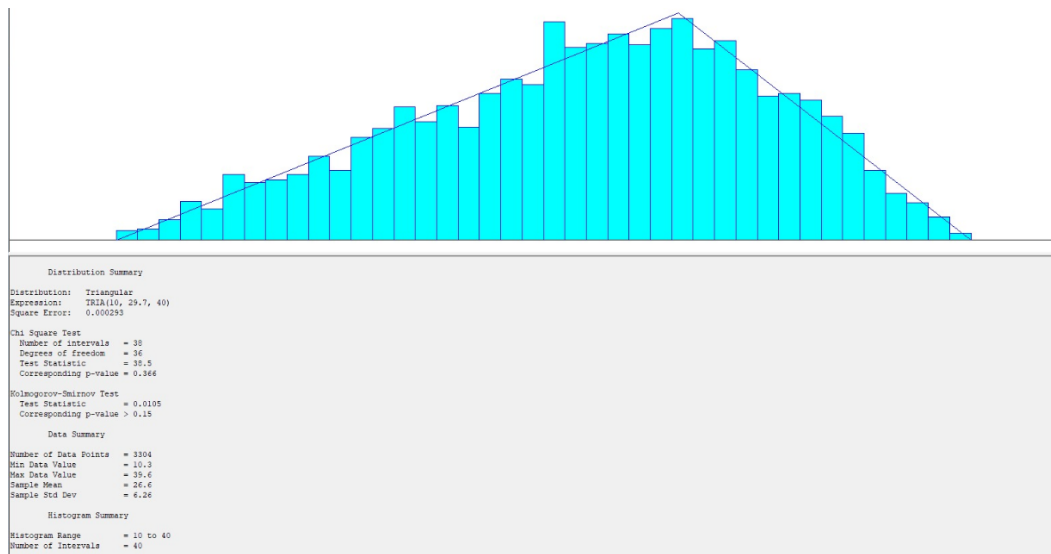


Figure C 4. Arena Input Analyzer result for the ID check process, where a triangular distribution was fitted to the observed data.

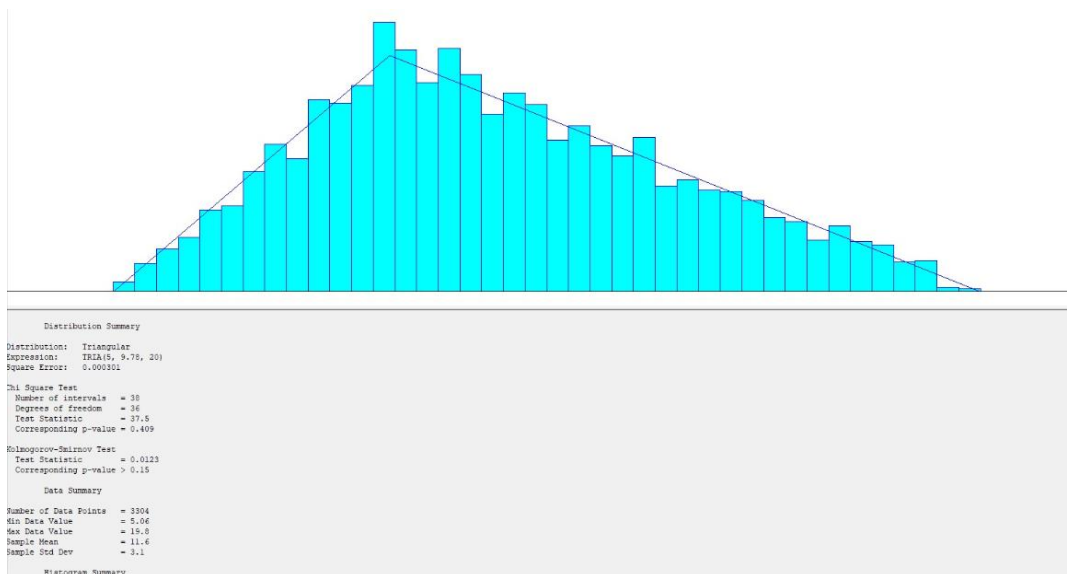


Figure C 5. Arena Input Analyzer result for the bag scan process. Since the theoretical distribution did not provide an acceptable fit, an empirical distribution was used in the simulation model.

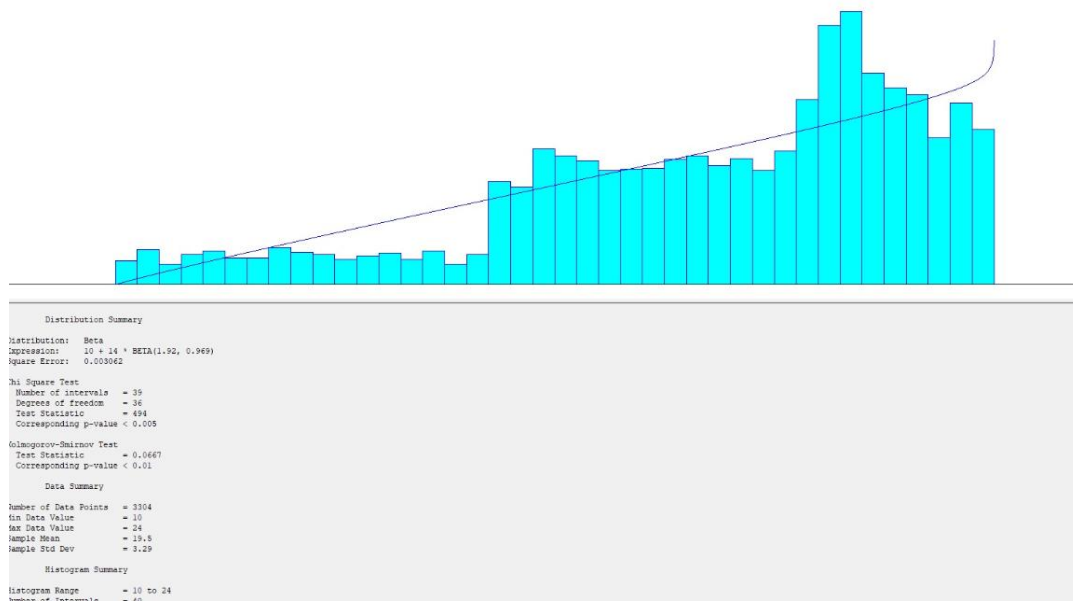


Figure C 6. Triangular distribution fit for the metal detector process using Arena Input Analyzer.

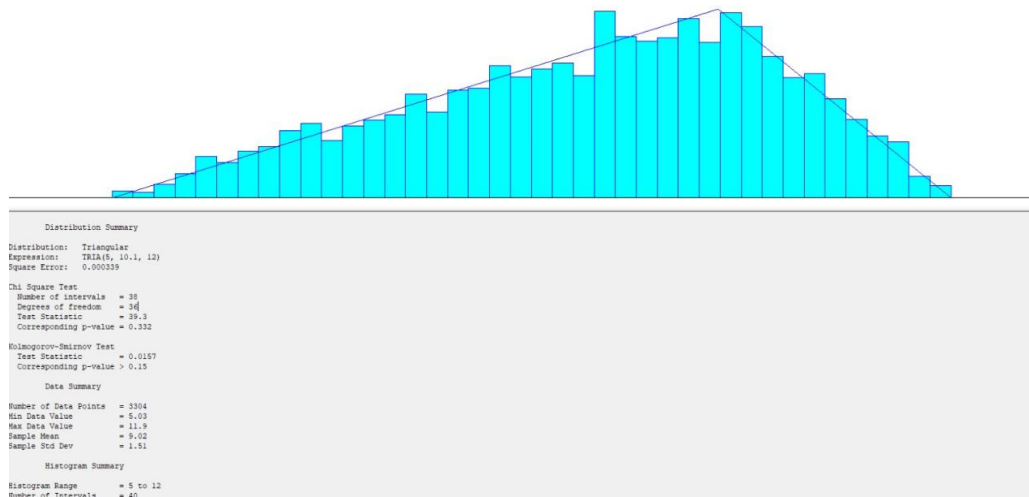


Figure C 7. Weibull distribution fit for the body search process using Arena Input Analyzer.

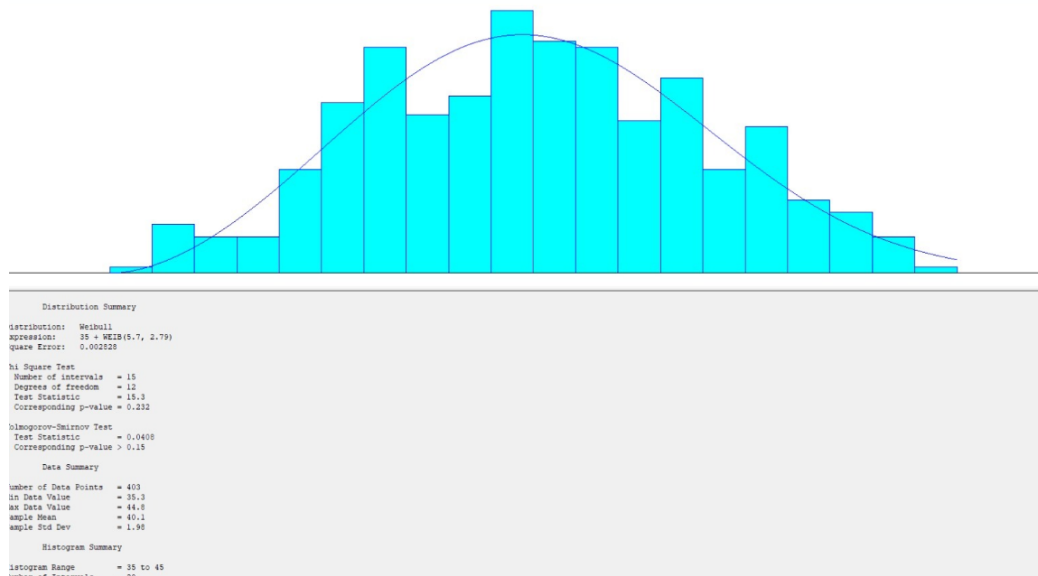


Figure C 8. Arena Input Analyzer result for the manual bag scan process, where a Weibull distribution was fitted to the observed data.

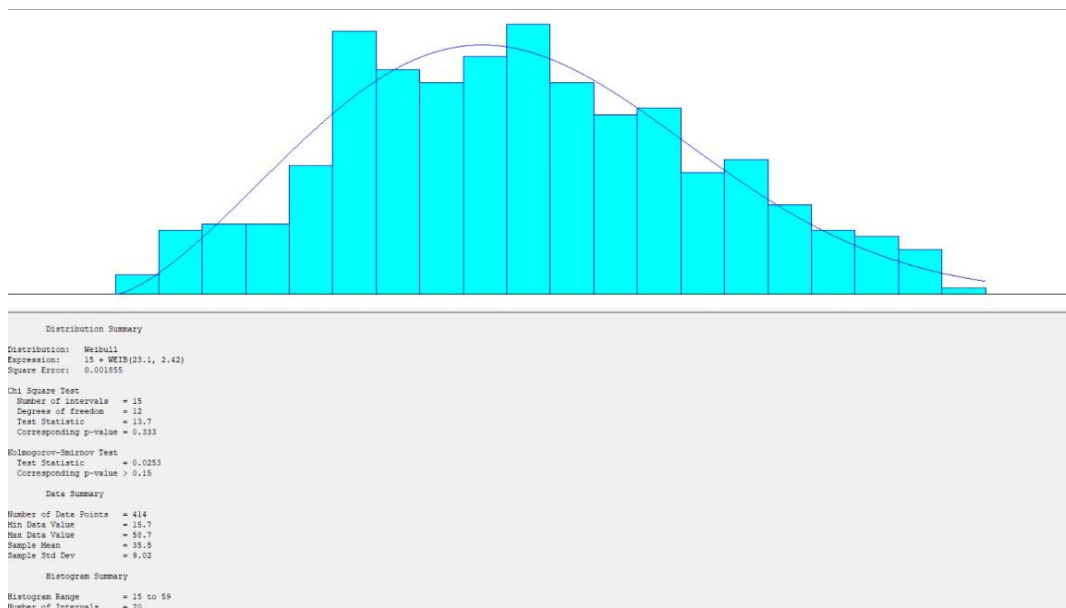


Figure C 9. Arena Input Analyzer result for the revesting process, where a triangular distribution was fitted to the observed data.

